CADT

Visualization Technique

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Data Visualization

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I. Definition

Data visualization is the graphical representation of information and data. **The main goal** of data visualization is to make data **easier to understand**, helping users **quickly spot patterns**, **trends, and outliers**. This is especially important with big data, where the large amount of information can be overwhelming without clear visuals. [1]

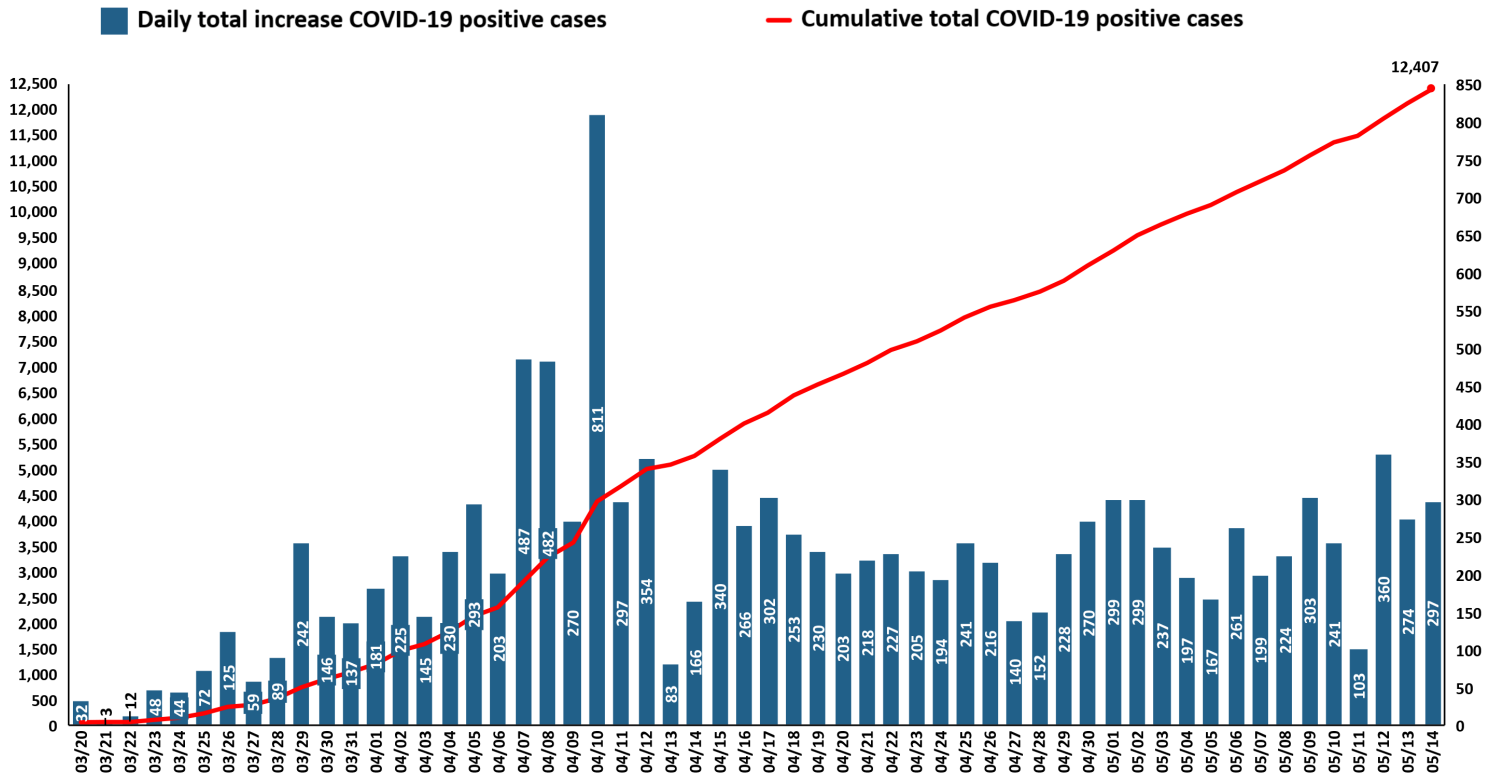


I. Why it is useful?

May 14, 2020

COVID-19 POSITIVE CASE GROWTH TREND

Greater Houston Area[1]



1. Austin, Brazoria, Chambers, Fort Bend, Galveston, Harris, Liberty, Montgomery and Waller

Source: USAFacts.org (https://usafacts.org/visualizations/coronavirus-covid-19-spread-map/)

TMC | TEXAS MEDICAL CENTER

"TMC" refers to the group of individual hospitals and institutions that make up Texas Medical Center

Monitoring threshold:

5-day trend of upward trajectory of documented cases

Current status:

Monitoring 1 day of case volume growth

This document is solely intended to share insights and best practices rather than specific recommendations. Individual institution data is shown as reported and has not been independently verified

I. Why it is useful?

Turn your data into a beautiful report or dashboard with **Pivot Tables**!

Order	Order Da	Customer	Customer Nan	Address	City
1015	01/27/14	27	Company AA	789 27th Street	Las V
1015	01/27/14	27	Company AA	789 27th Street	Las V
1016	01/04/14	4	Company D	123 4th Street	New
1016	01/04/14	4	Company D	123 4th Street	New
1016	01/04/14	4	Company D	123 4th Street	New
1017	01/12/14	12	Company L	123 12th Street	Las V
1017	01/12/14	12	Company L	123 12th Street	Las V
1018	01/08/14	8	Company H	123 8th Street	Portl
1019	01/04/14	4	Company D	123 4th Stree	
1020	01/29/14	29	Company CC	789 29th Stre	
1021	01/03/14	3	Company C	123 3rd Stree	
1022	01/06/14	6	Company F	123 6th Stree	
1023	01/28/14	28	Company BB	789 28th Stre	
1024	01/08/14	8	Company H	123 8th Stree	
1025	01/10/14	10	Company J	123 10th Stre	
1026	01/07/14	7	Company G	123 7th Stree	
1027	01/10/14	10	Company J	123 10th Stre	
1027	01/10/14	10	Company J	123 10th Stre	
1027	01/10/14	10	Company J	123 10th Street	Chica
1028	01/11/14	11	Company K	123 11th Street	Mian
1028	01/11/14	11	Company K	123 11th Street	Mian
1029	01/01/14	1	Company A	123 1st Street	Seatt
1029	01/01/14	1	Company A	123 1st Street	Seatt
1029	01/01/14	1	Company A	123 1st Street	Seatt
1030	01/28/14	28	Company BB	789 28th Street	Mem
1030	01/28/14	28	Company BB	789 28th Street	Mem
1031	01/09/14	9	Company I	123 9th Street	Salt L
1031	01/09/14	9	Company I	123 9th Street	Salt L
1032	01/06/14	6	Company F	123 6th Street	Milw
1033	02/08/14	8	Company H	123 8th Street	Portl
1034	02/03/14	3	Company C	123 3rd Street	Los A
1034	02/03/14	3	Company C	123 3rd Street	Los A



I. Why Visualization is Critical in Data Science

Would you prefer to **view large data** tables and then make sense of that data or view a data visualization that represents that data in an easy-to-understand visual format?

- **It helps explore data:** Charts and graphs make it easier to see **patterns**, **trends**, and unusual values that are hard to notice in tables.

- **It communicates insights clearly:** Visuals help explain **complex** data in a simple way so even non-technical people can understand.

I. Why Visualization is Critical in Data Science

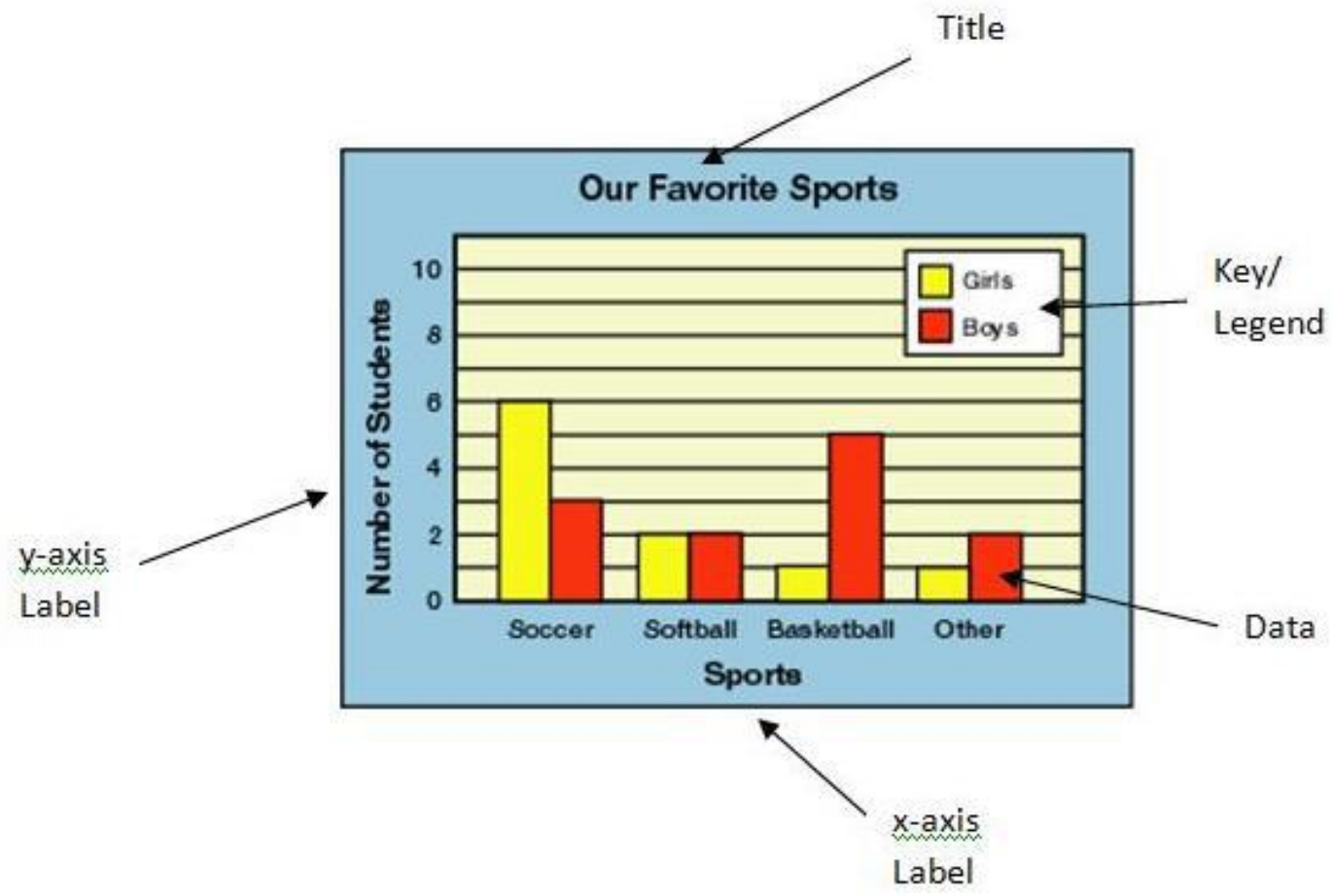
Would you prefer to **view large data** tables and then make sense of that data or view a data visualization that represents that data in an easy-to-understand visual format?

- **It supports decision-making:** When data is shown visually, stakeholders can quickly understand the situation and make better decisions.

- **It helps tell a story:** Good visualizations create a clear and meaningful story from the data, making the message more powerful and engaging.

I. Type of chart

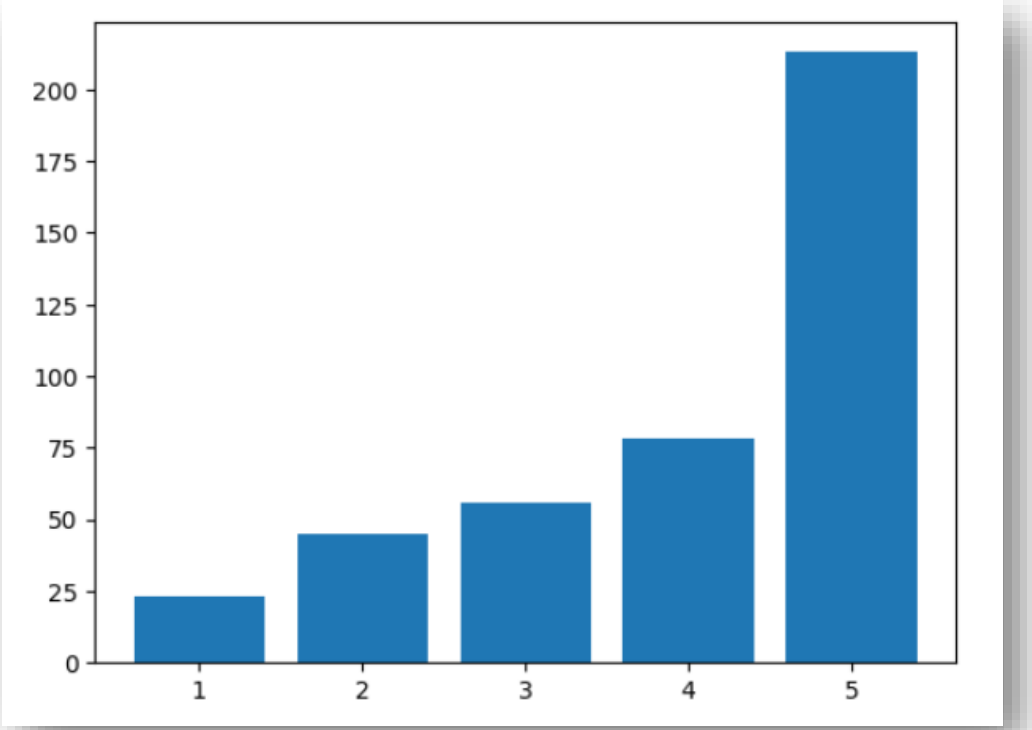
- Charts typically consist of several key components:
  - Axes labels
  - Title
  - Legends
  - Data points



I. Type of chart

- Bar chart
  - Used to **compare values across different categories**
  - Each category is represented by a rectangular bar, and the **length or height of the bar shows the value.**

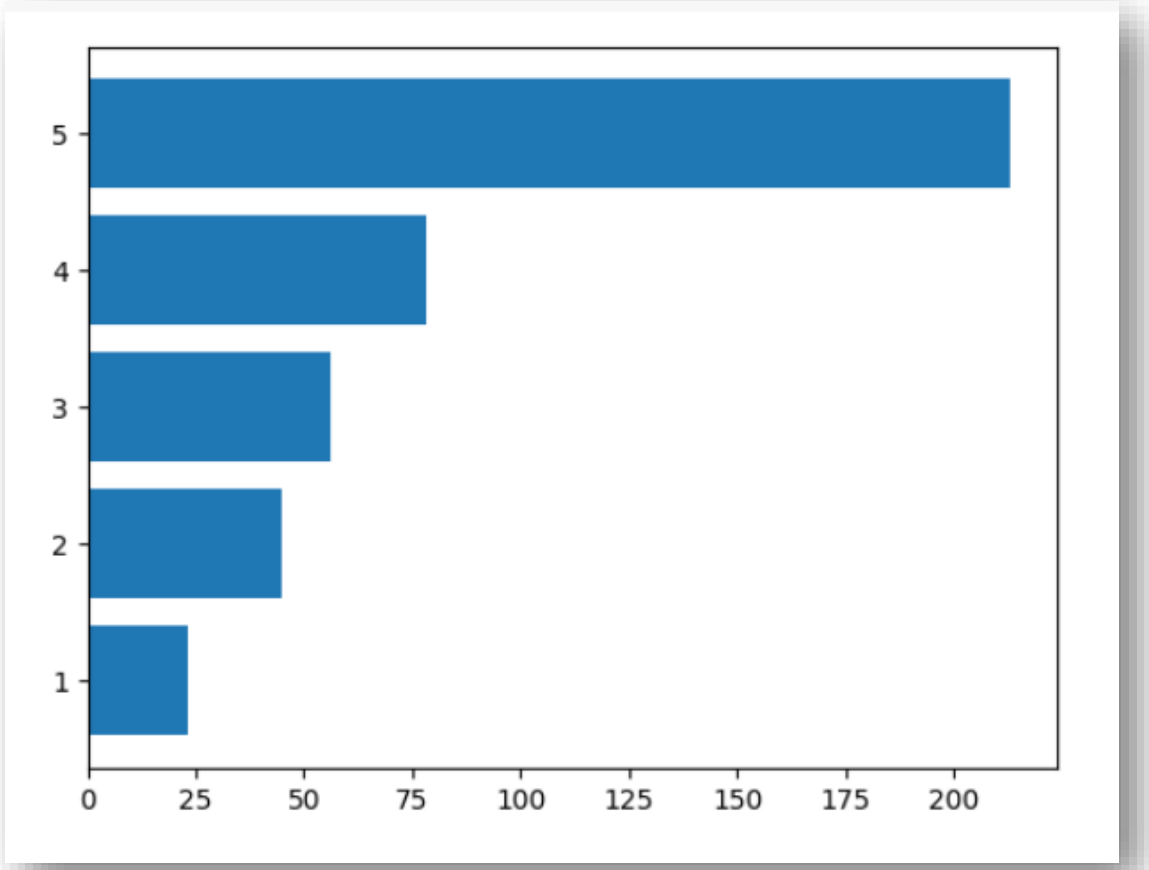
Vertical Bar Graph



I. Type of chart

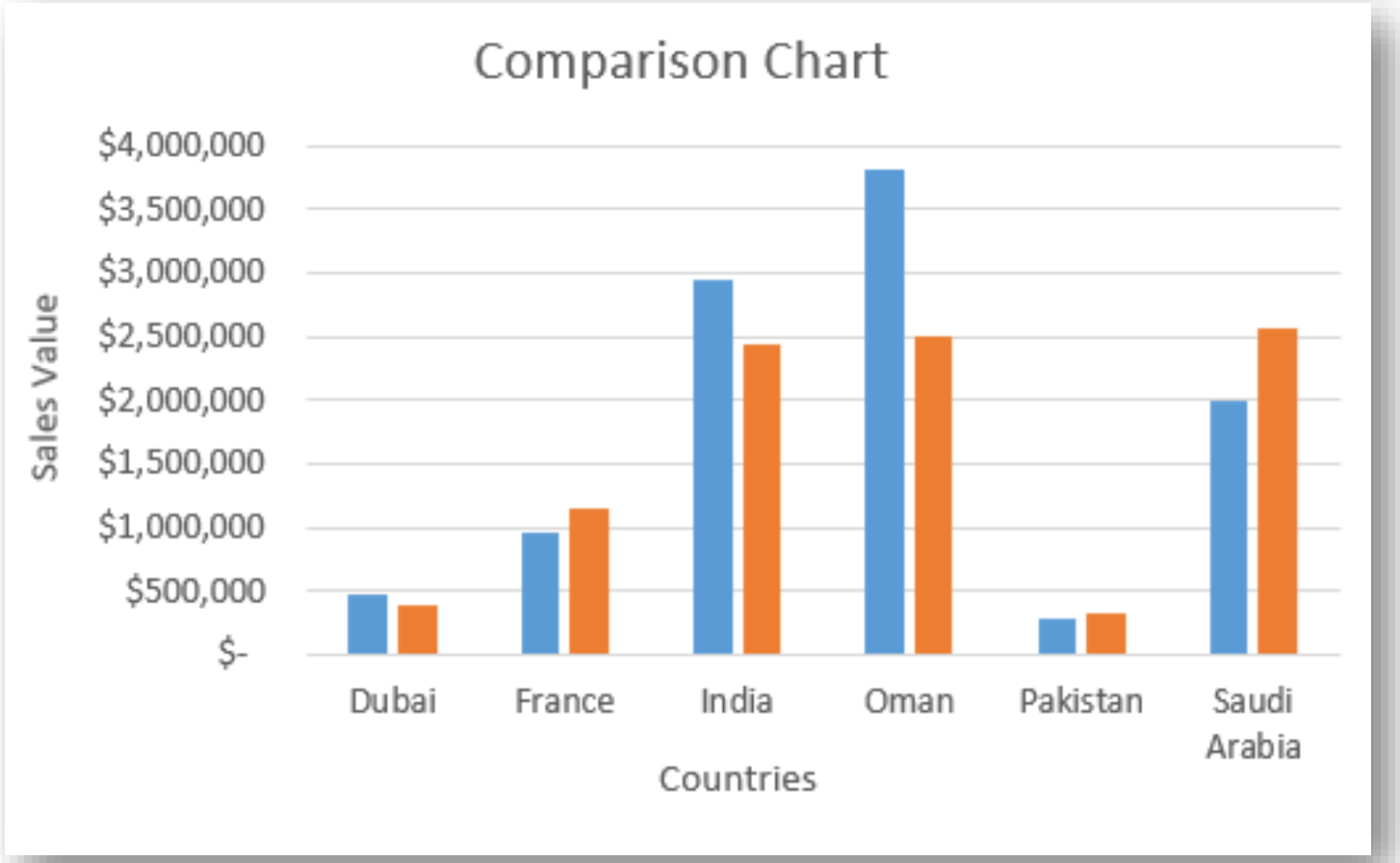
A horizontal bar graph is used to compare and **rank** categories, especially when category names **are long or there are many categories**.

Horizontal Bar Graph.



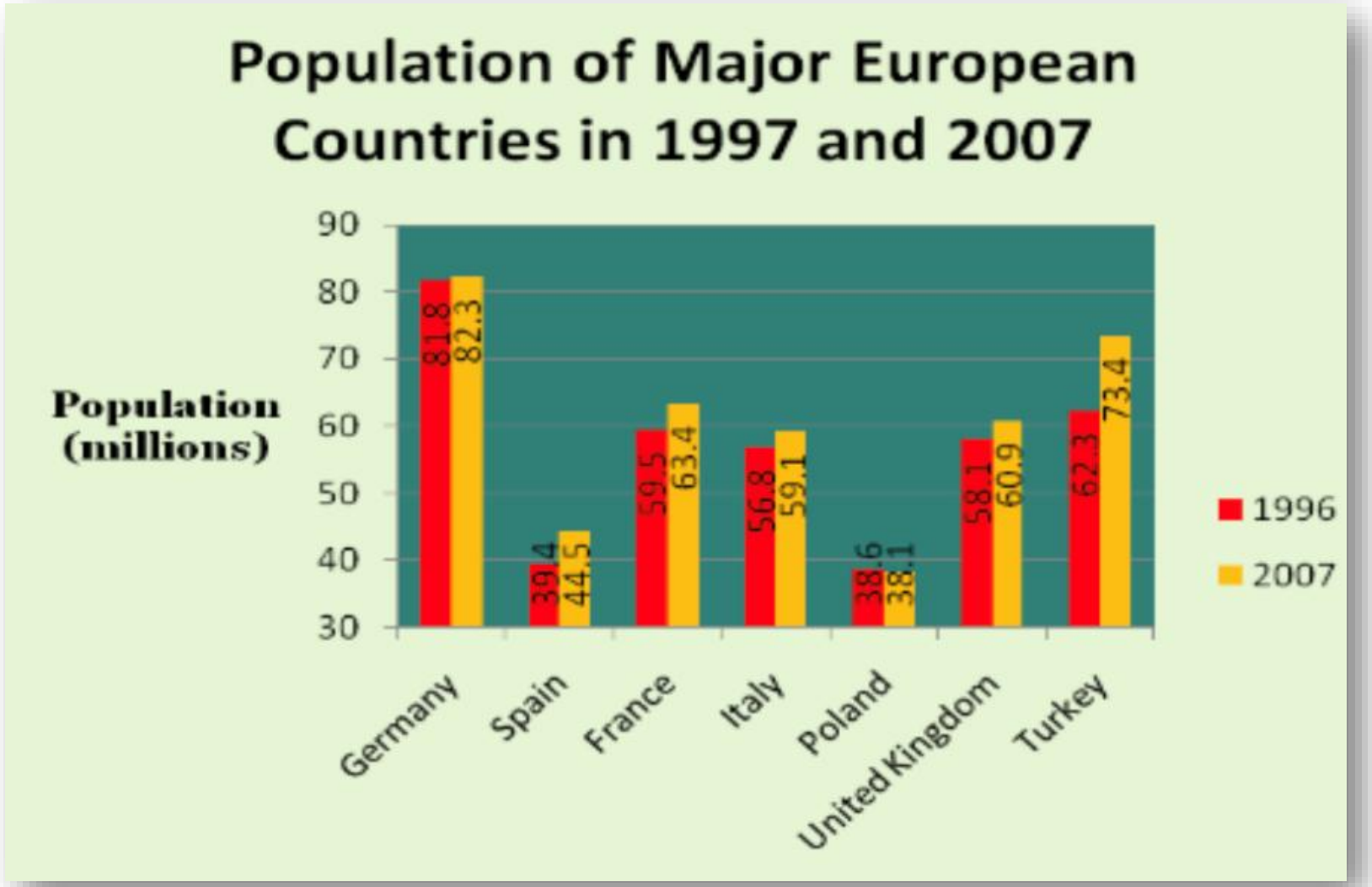
I. Type of chart

Grouped bar graphs are used when the datasets have subgroups that need to be visualized on the graph. The subgroups are differentiated by distinct colors.



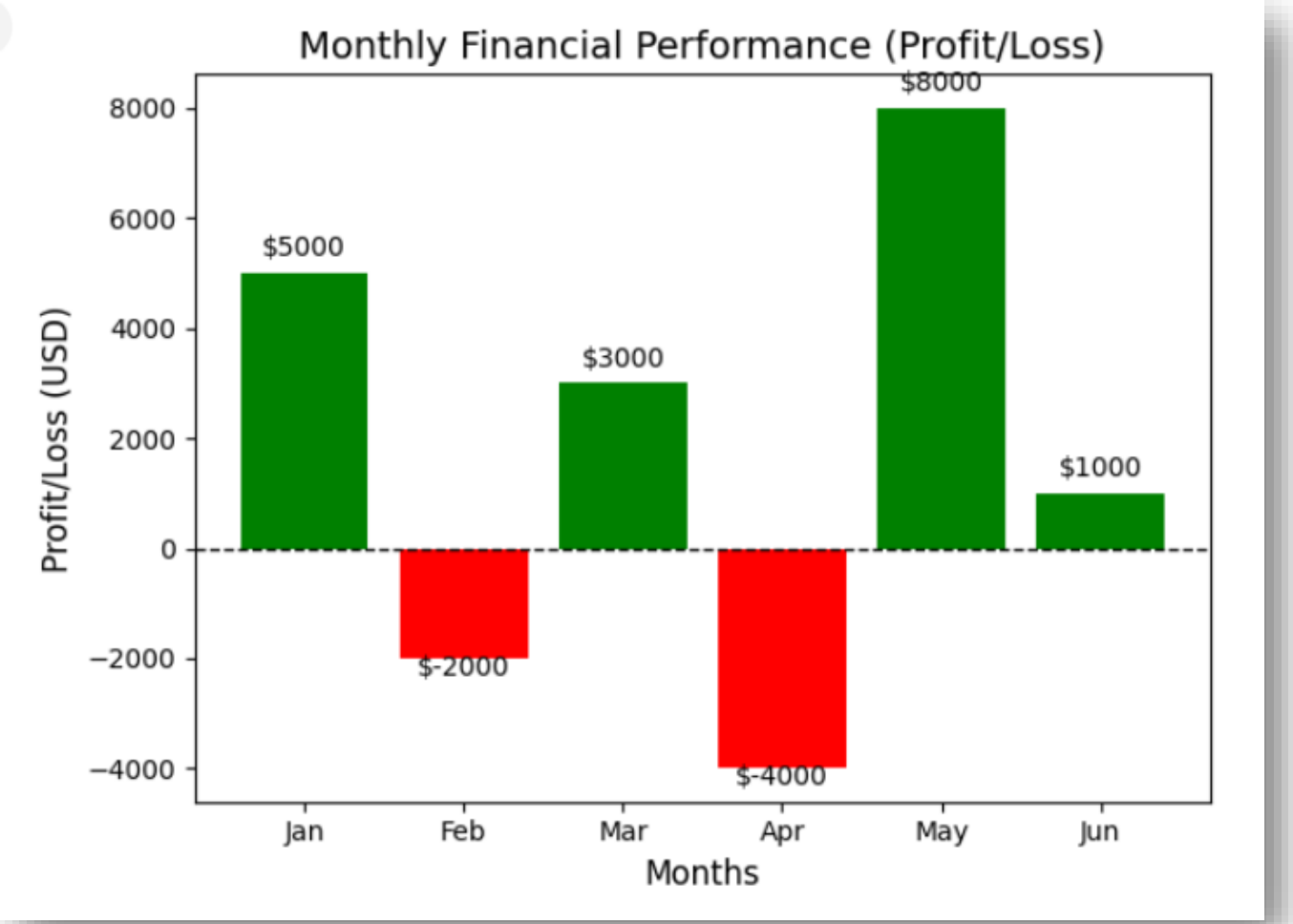
I. Type of chart

Percentage



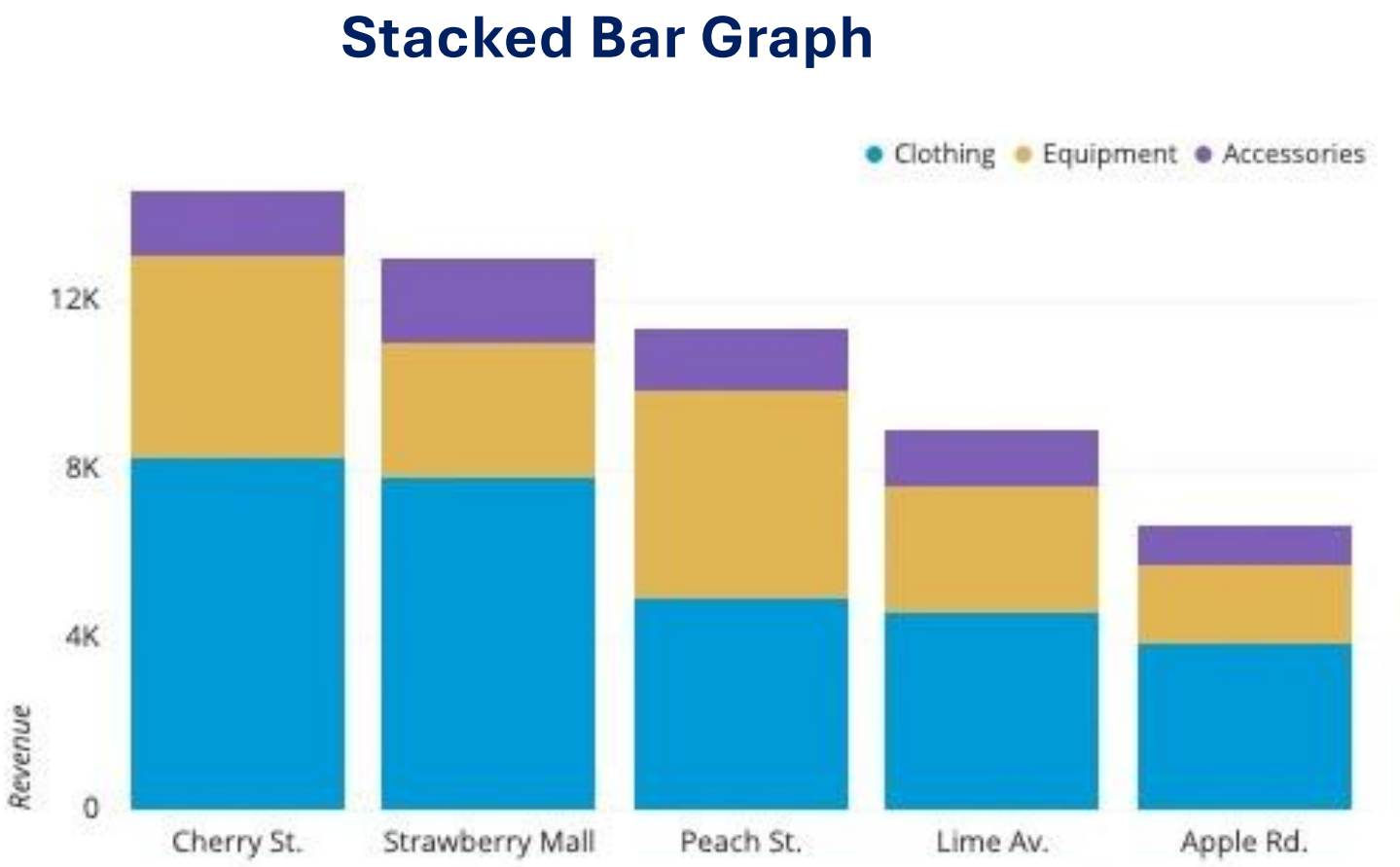
I. Type of chart

Deviation



I. Type of chart

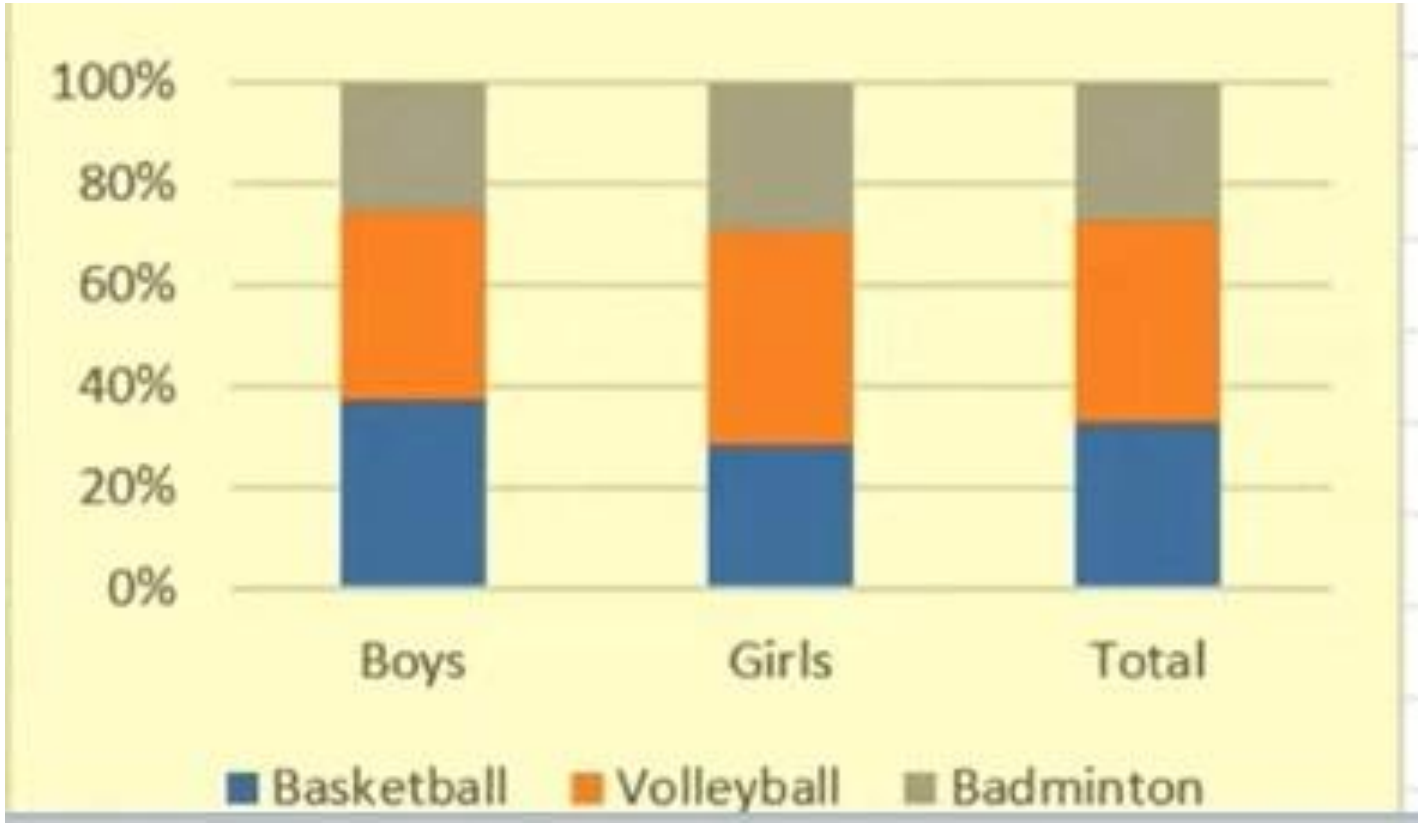
The stacked bar graphs are used to show dataset subgroups.



I. Type of chart

This is the type of stacked bar graph where each stacked bar shows the percentage of its discrete value from the total value. The total percentage is 100%.

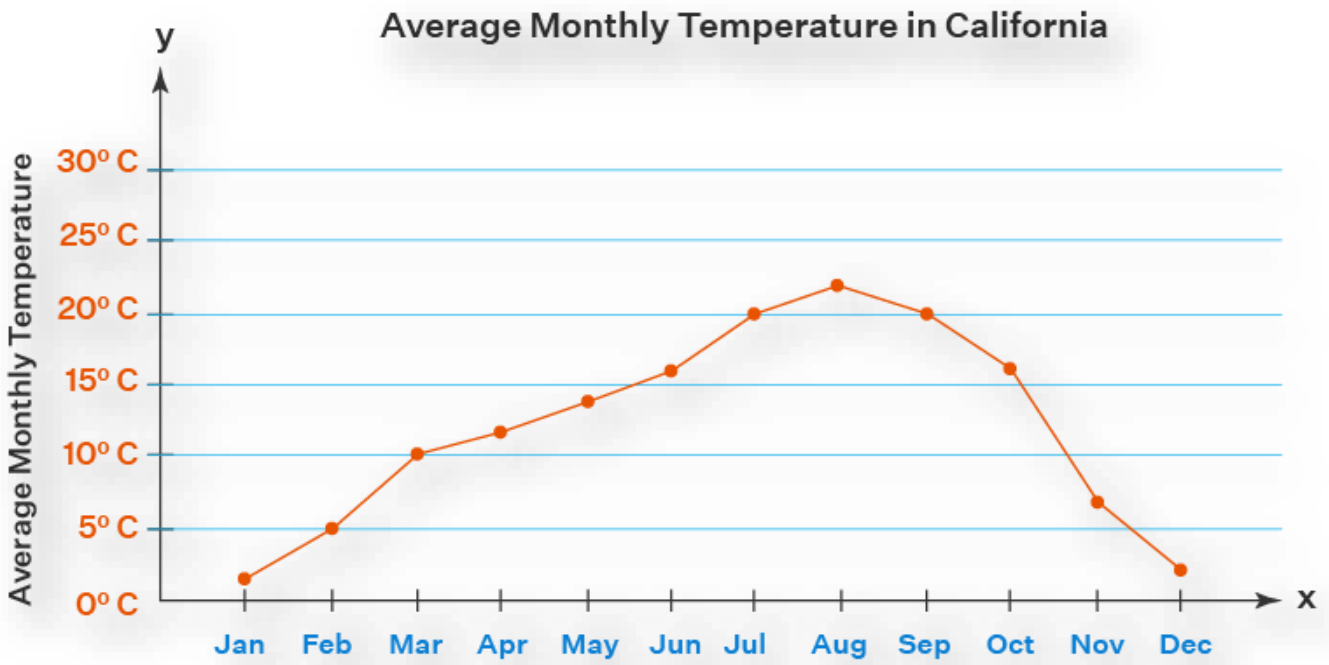
Segmented Bar Graph



I. Type of chart

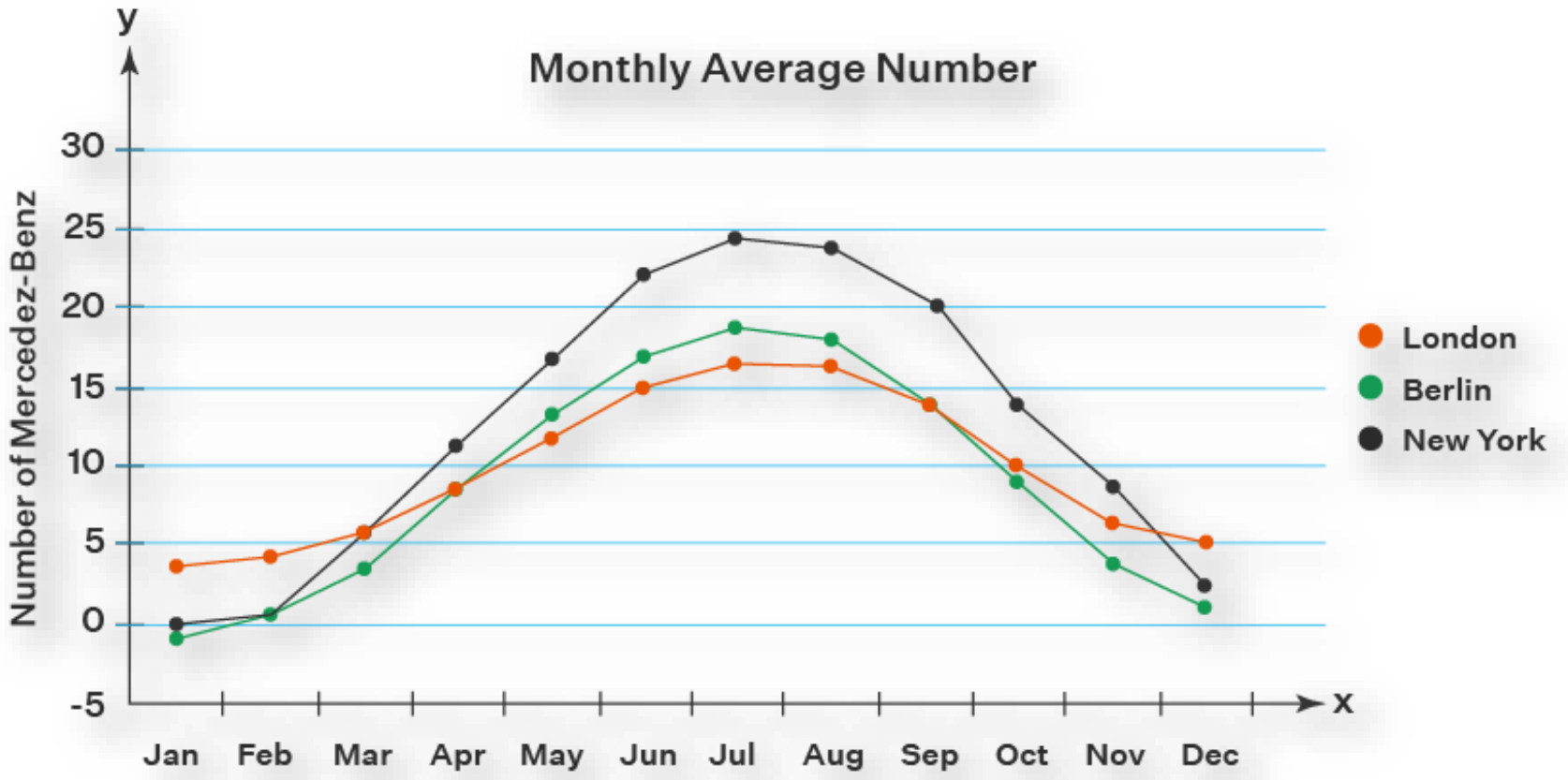
Line Graph is Line Graph is a sequence of data points as markers. The points are ordered typically by their x-axis value. These points are joined with straight line segments. A line graph is used to visualize a trend in data over intervals of time.

Simple line chart



I. Type of chart

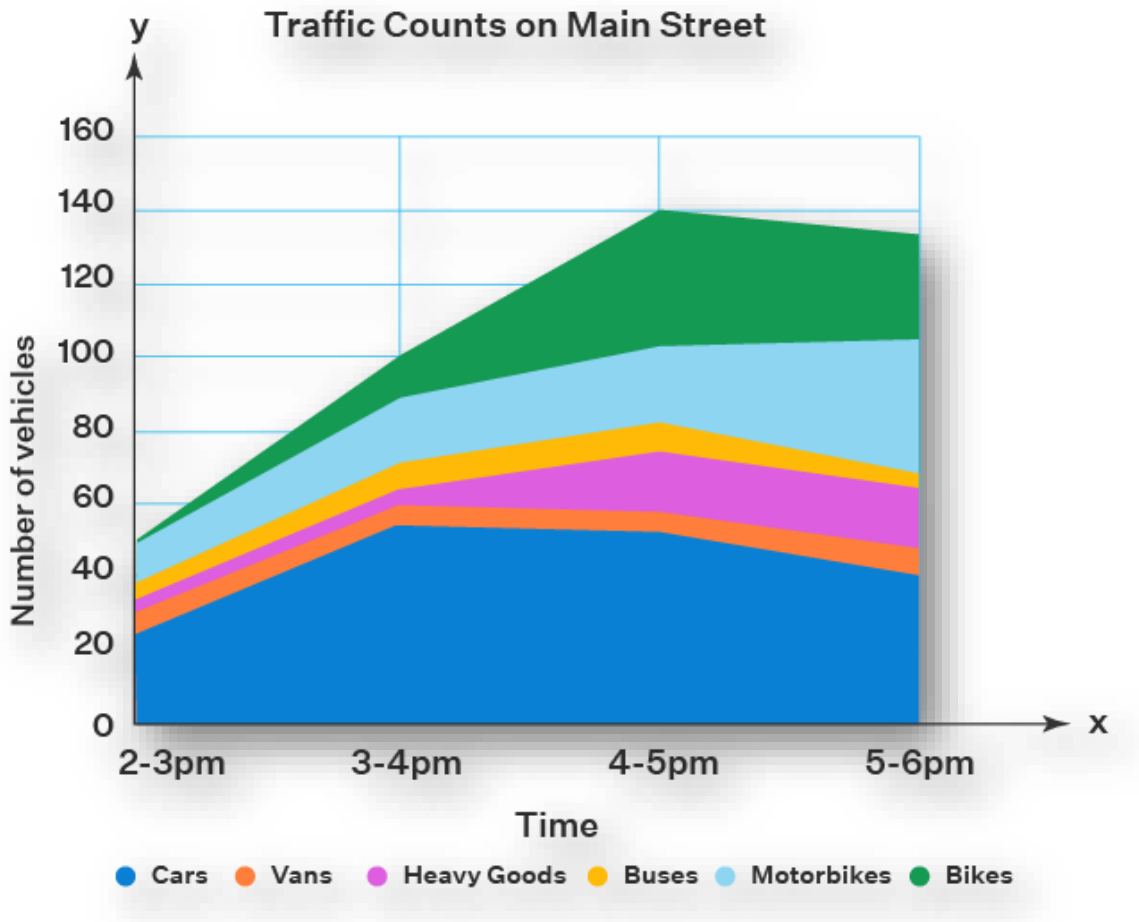
Multiple Line Chart



I. Type of chart

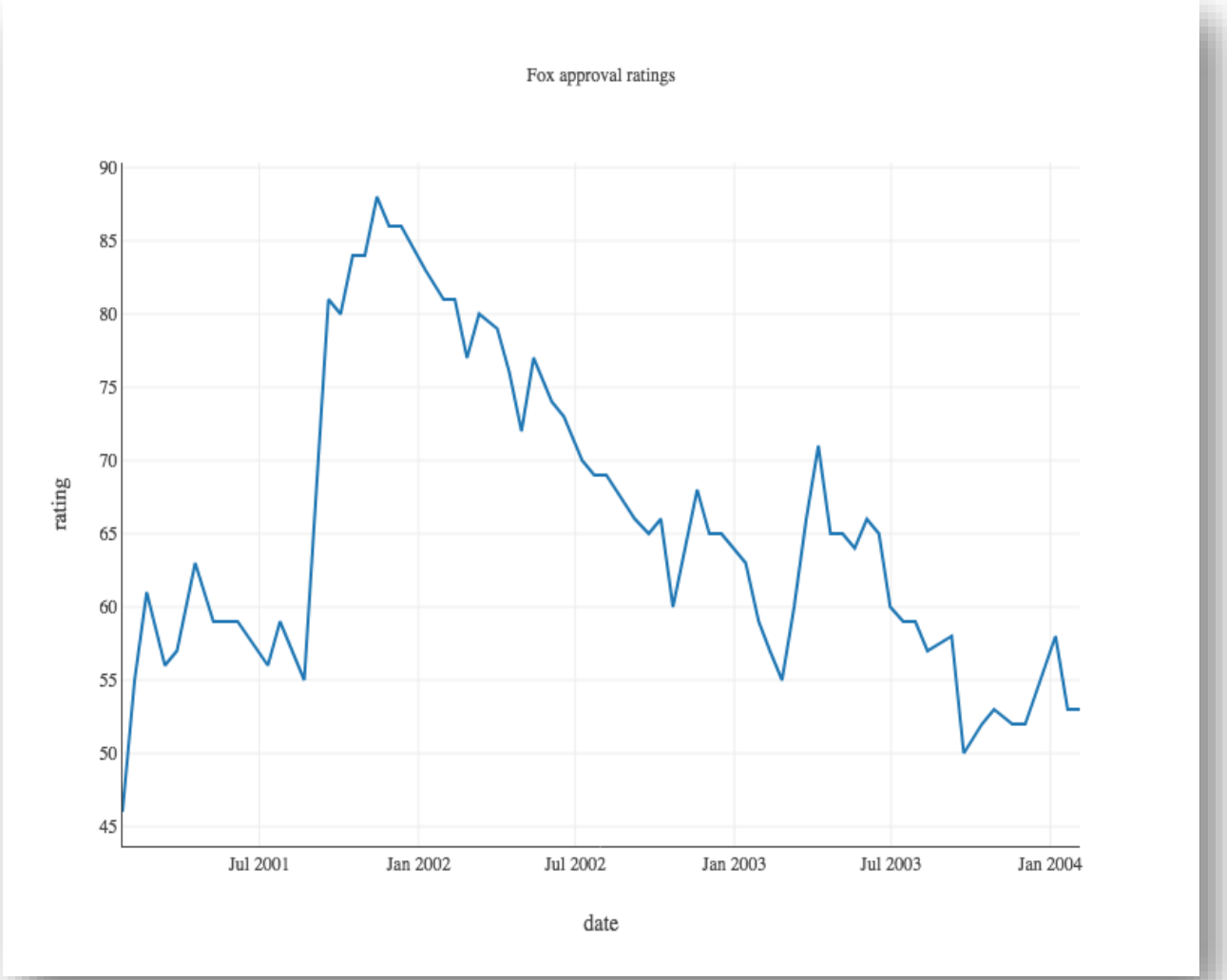
Compare Stacked **Area** Chart

Compound Line Chart



I. Type of chart

Time series



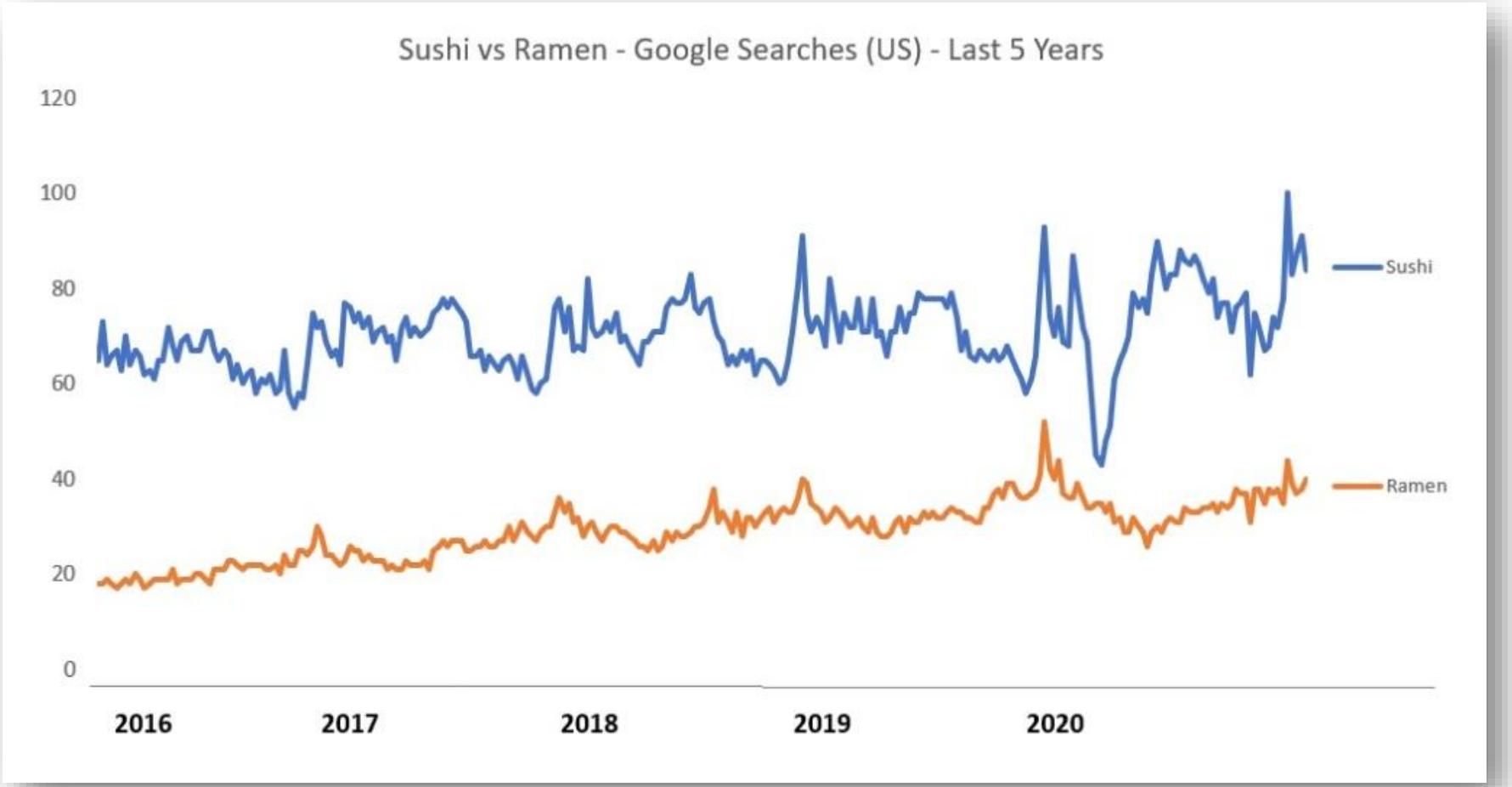
I. Type of chart

Trends & Pattern



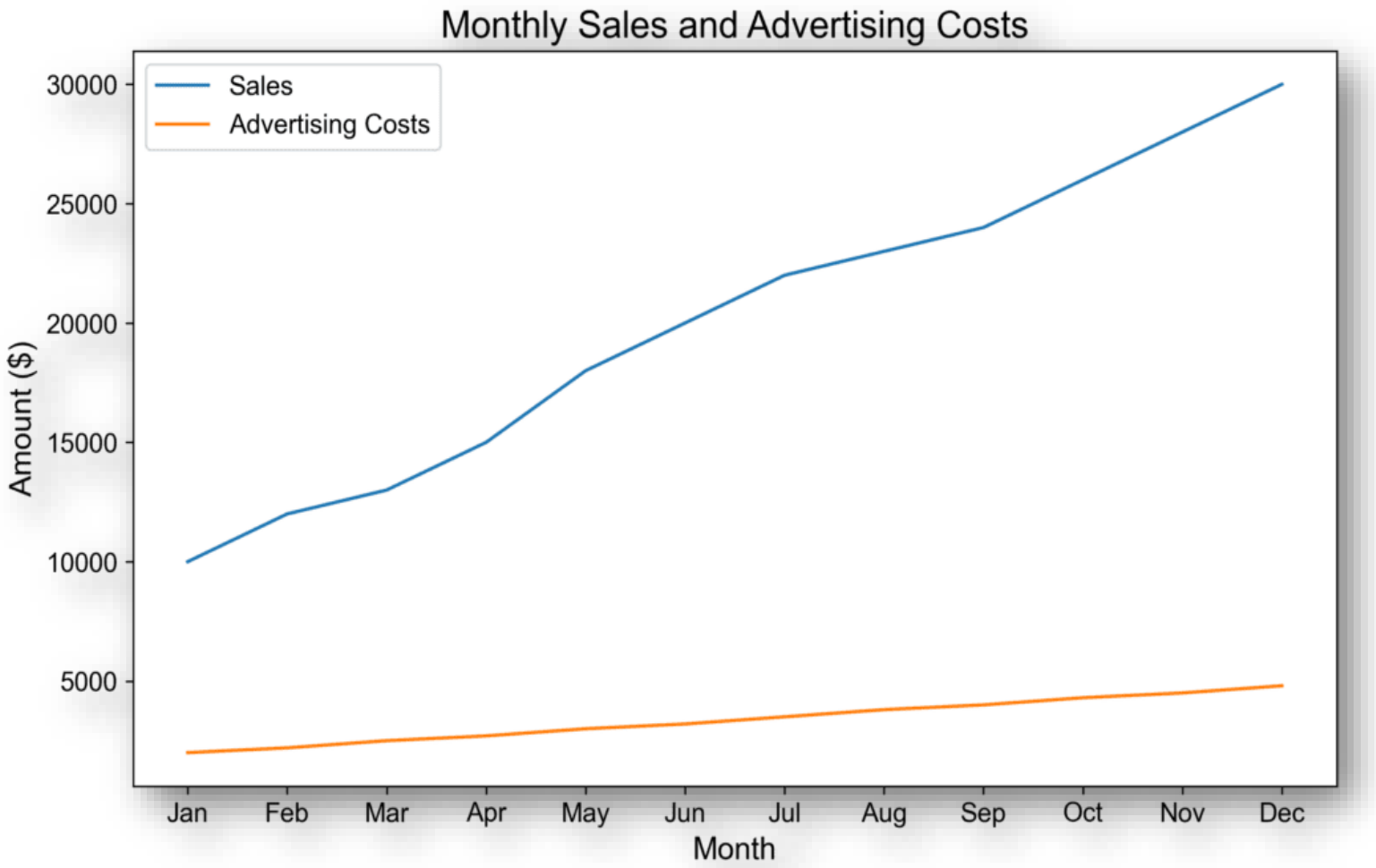
I. Type of chart

Comparison



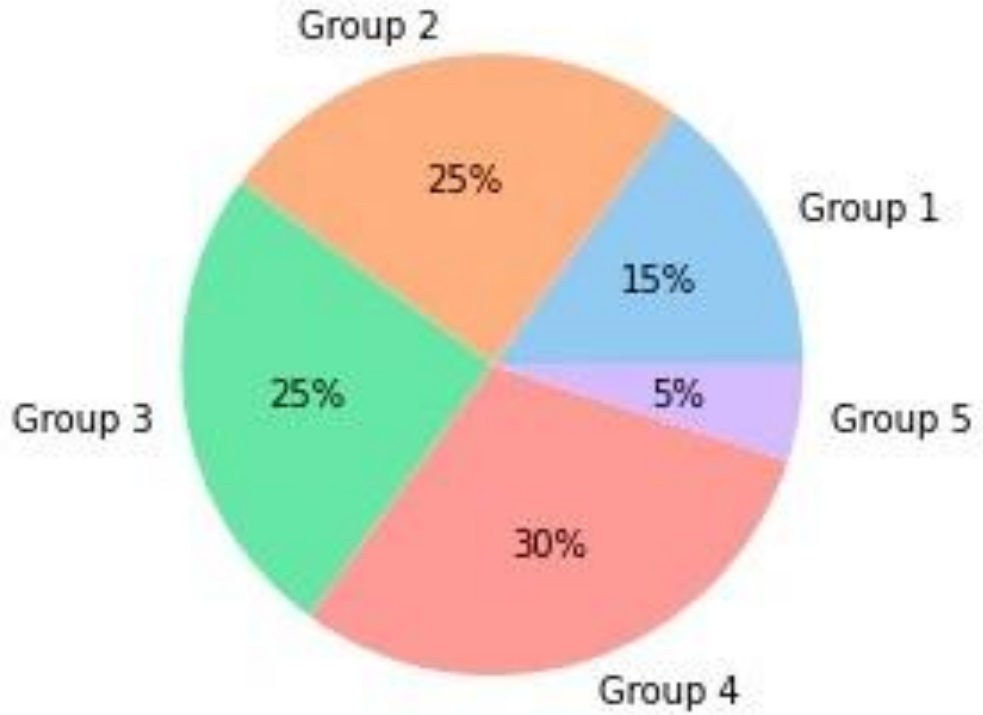
I. Type of chart

Correlation

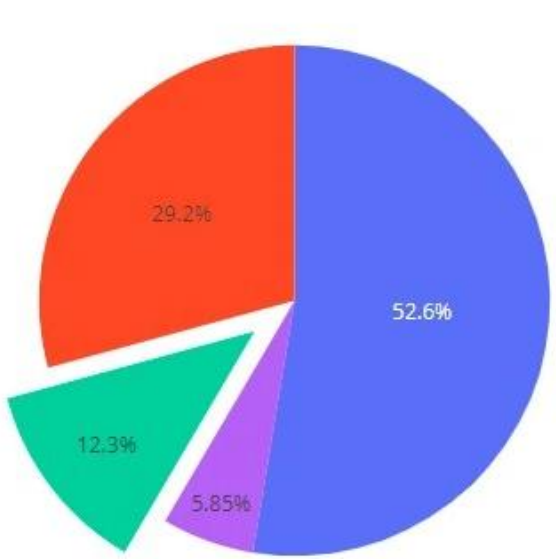


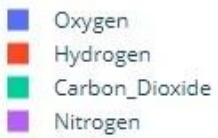
I. Type of chart

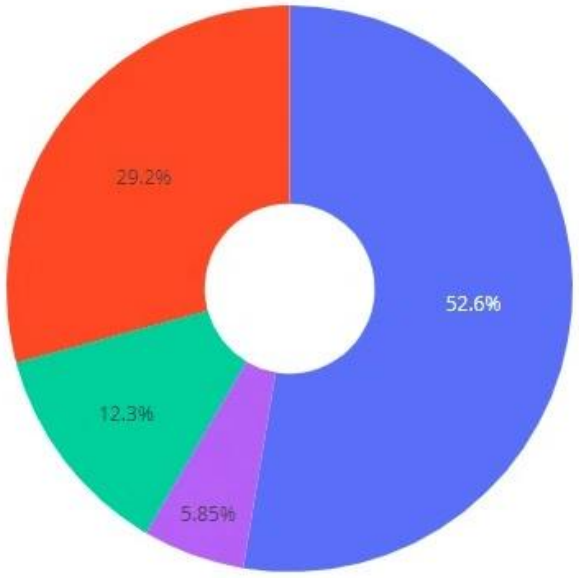
A pie chart is a circular statistical graphic. To illustrate numerical proportion, it is divided into slices. In a pie chart, for every slice, each of its arc lengths is proportional to the amount it represents. The central angles, and area are also proportional. It is named after a sliced pie.

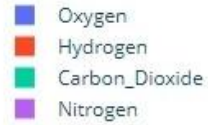


I. Type of chart









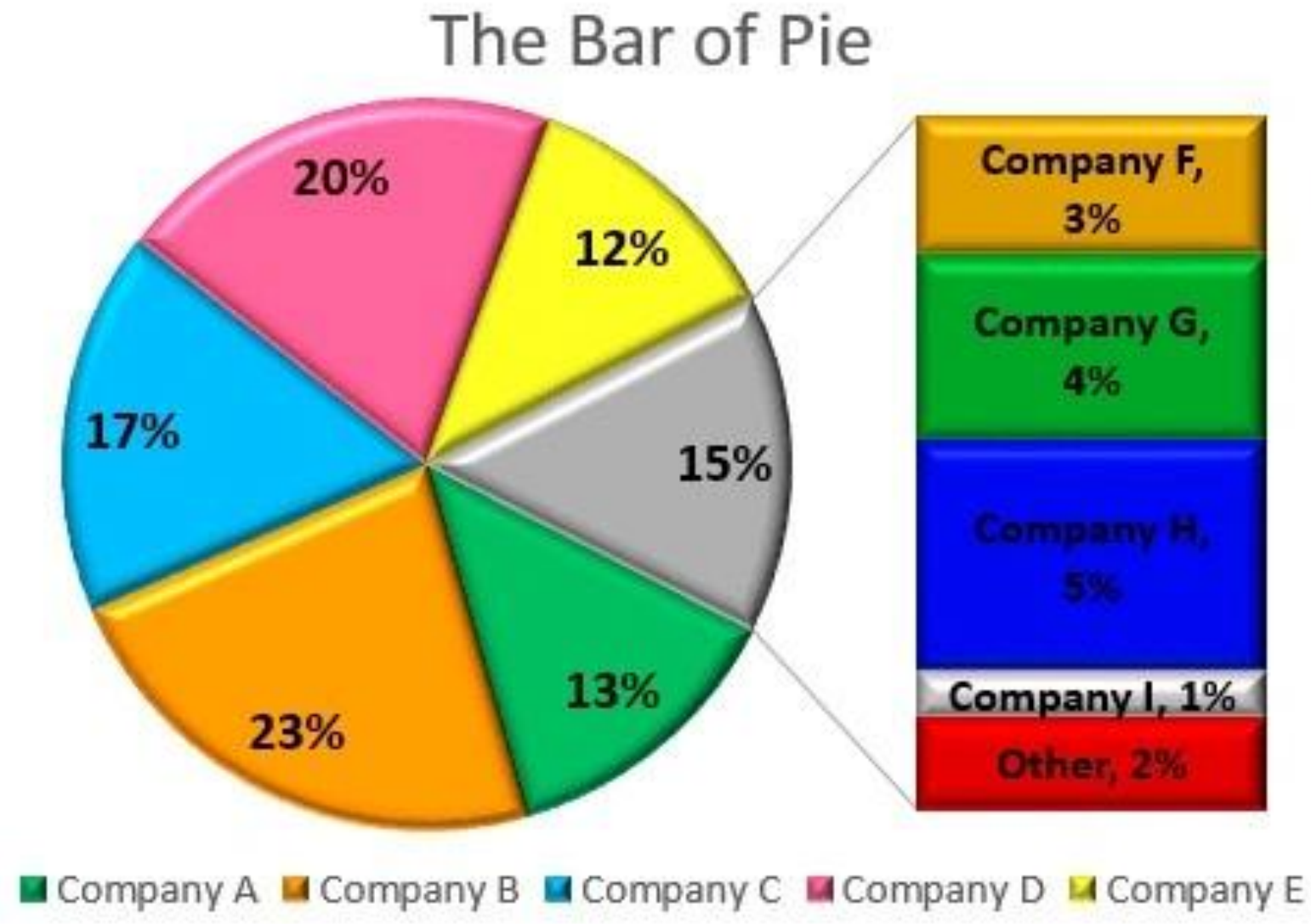
Data Visualization

I. Type of chart

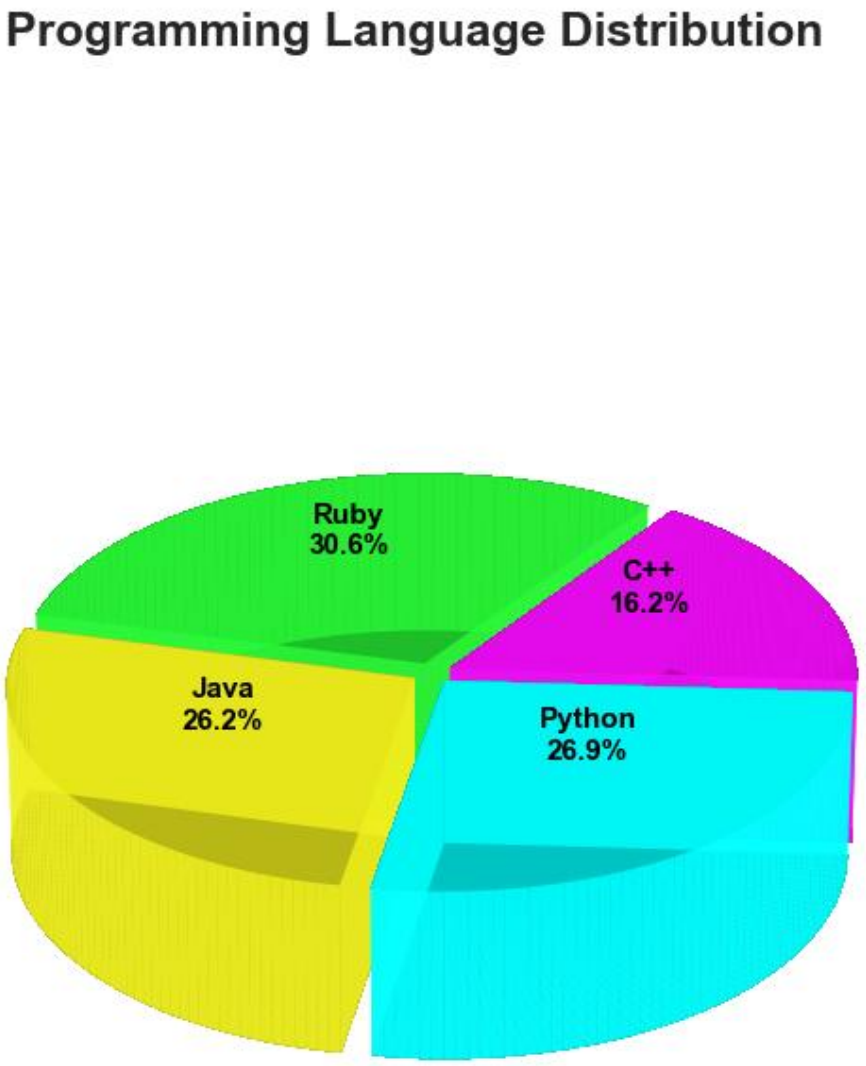
A pie of pie is a chart that generates an entirely new pie chart detailing a small sector of the existing pie chart. It can be used to reduce the clutter and emphasize a particular group of elements.



I. Type of chart



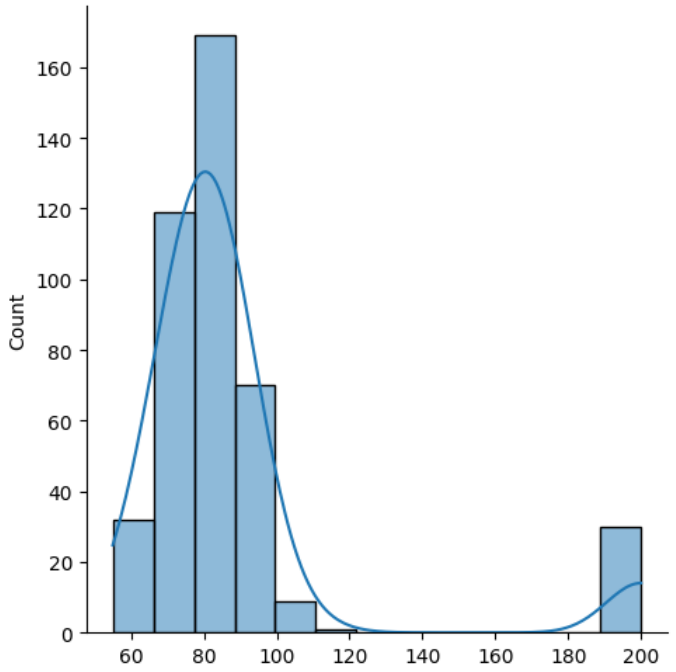
I. Type of chart

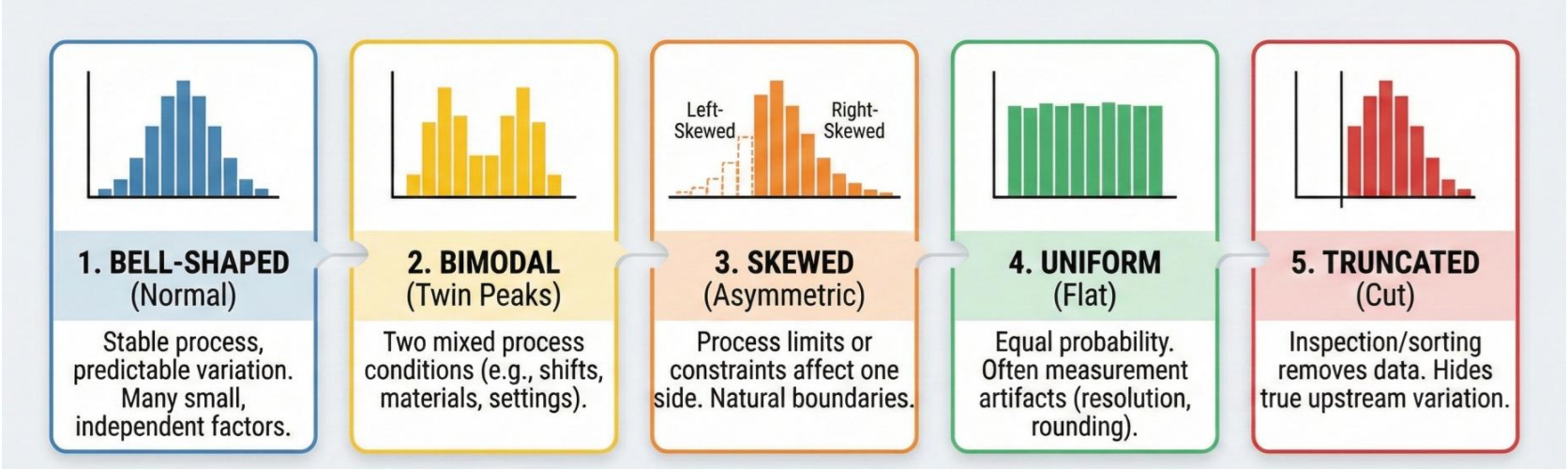


I. Type of chart

Histogram: Distribution of a variable

- See the **frequency of values** in different ranges (bins).
- **Example:** Heights of students grouped into ranges like 150–155 cm, 155–160 cm, etc.
- Helps identify patterns like skewness, peaks, or outliers in the data.



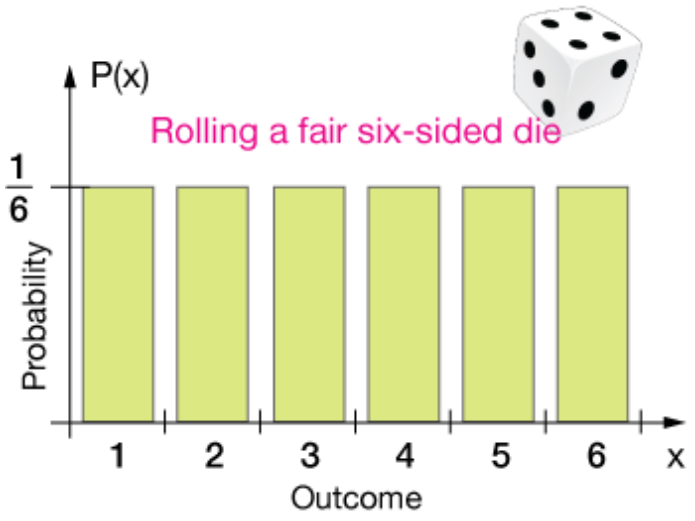


I. Histogram

- A distribution is a function that shows the **possible values** for a variable and **how often** they occur.

Probability distribution (one dice)

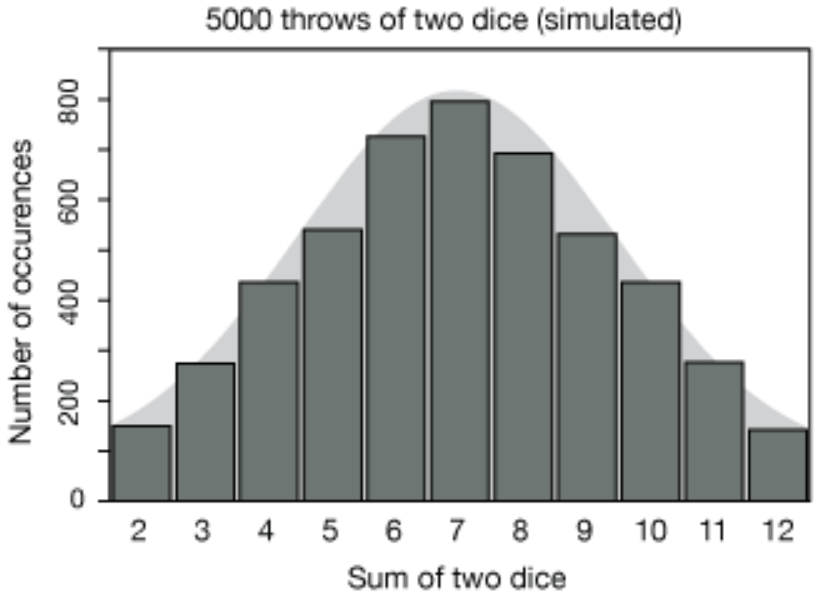
Value (X)	Probability P(X)
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6



Probability distribution (two dice)

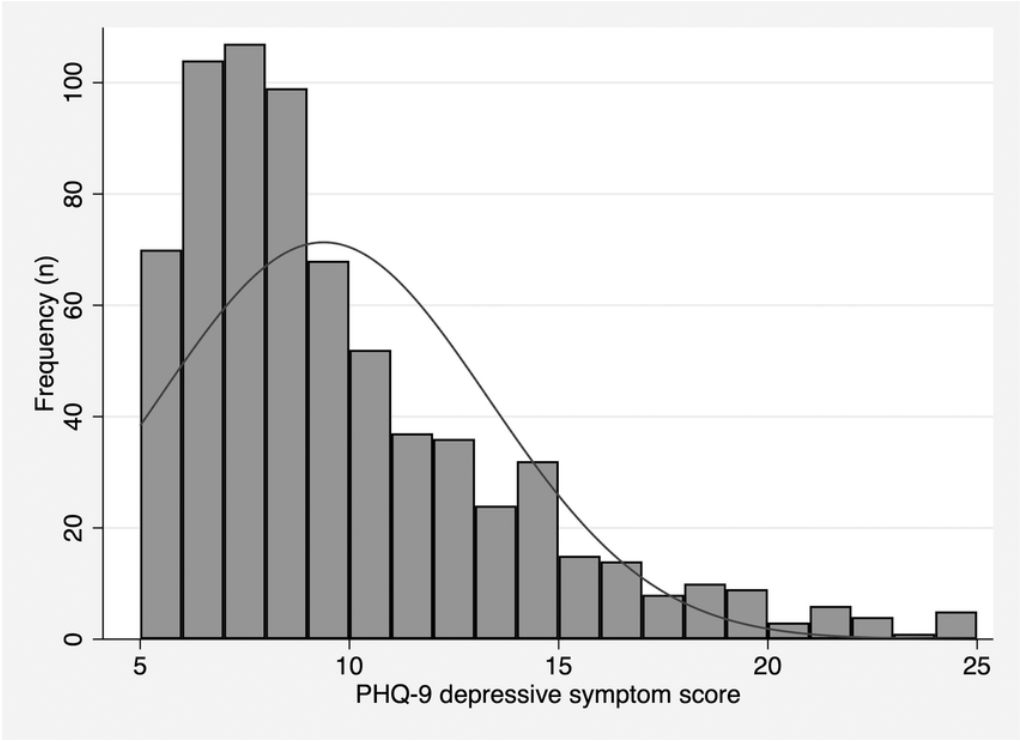
Sum (X)	Probability P(X)
2	1/36
3	2/36
4	3/36
5	4/36
6	5/36
7	6/36
8	5/36
9	4/36
10	3/36
11	2/36
12	1/36

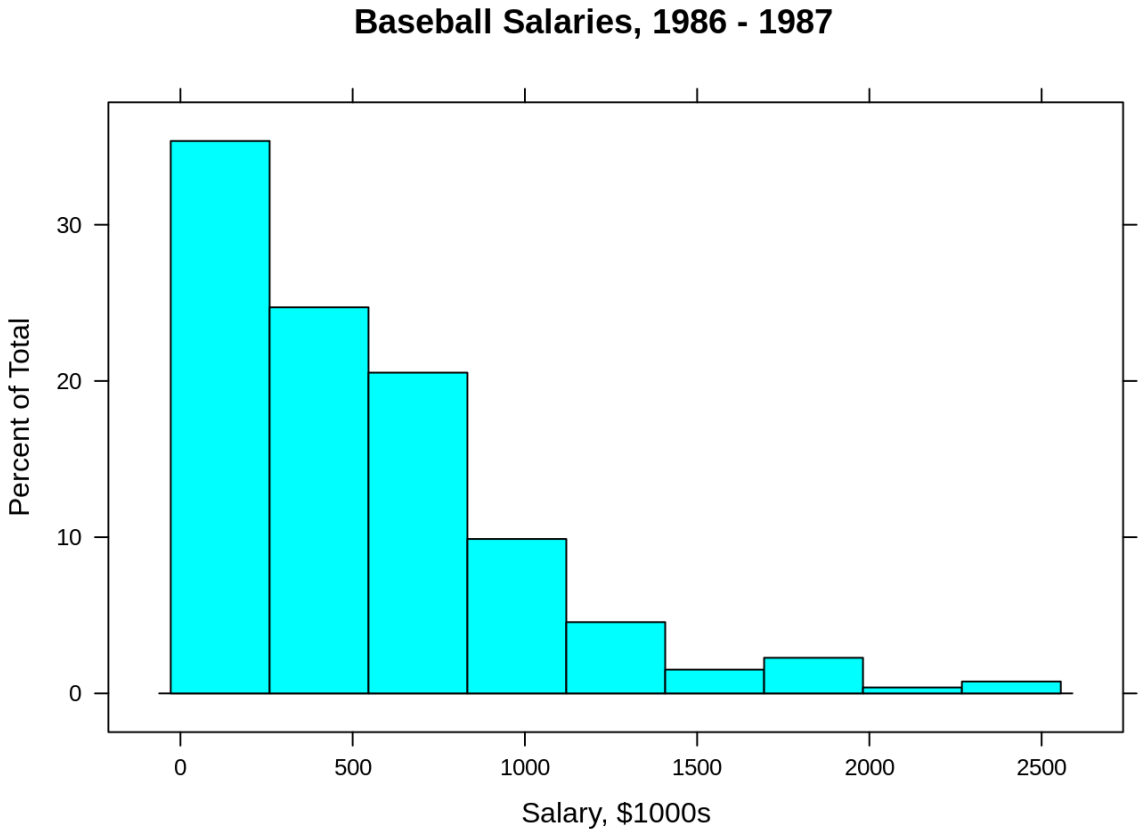




I. Type of chart

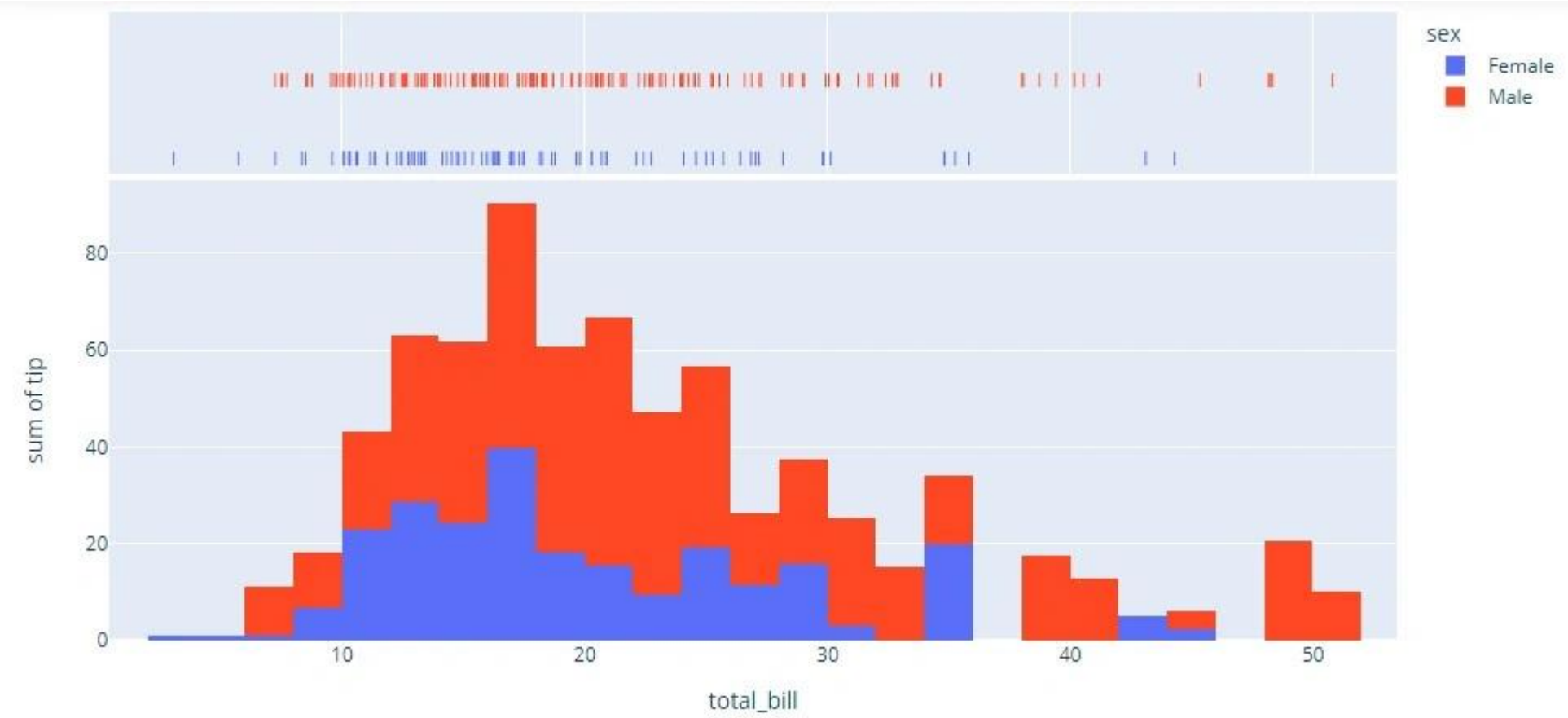
Histogram: Distribution of a variable





I. Type of chart

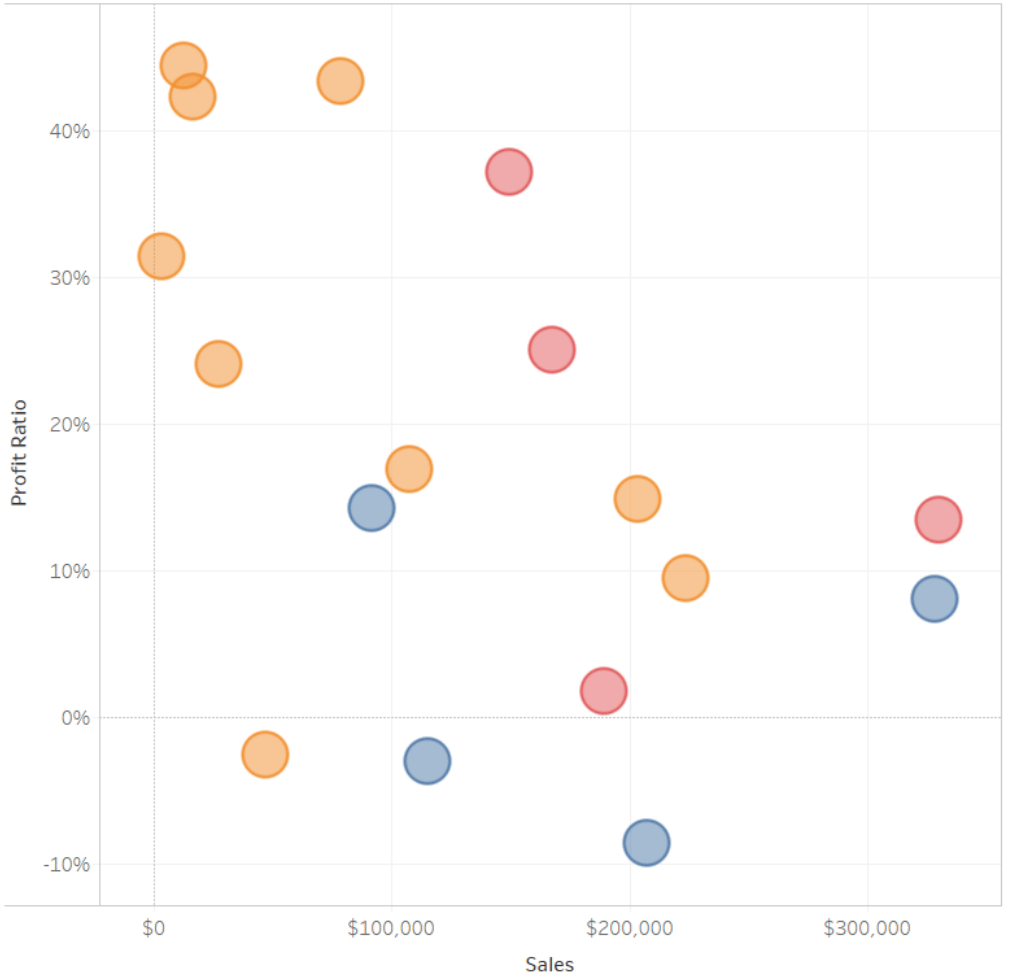
Histogram: Distribution of a variable



I. Type of chart

Scatter plot: Relationship between two numeric variables

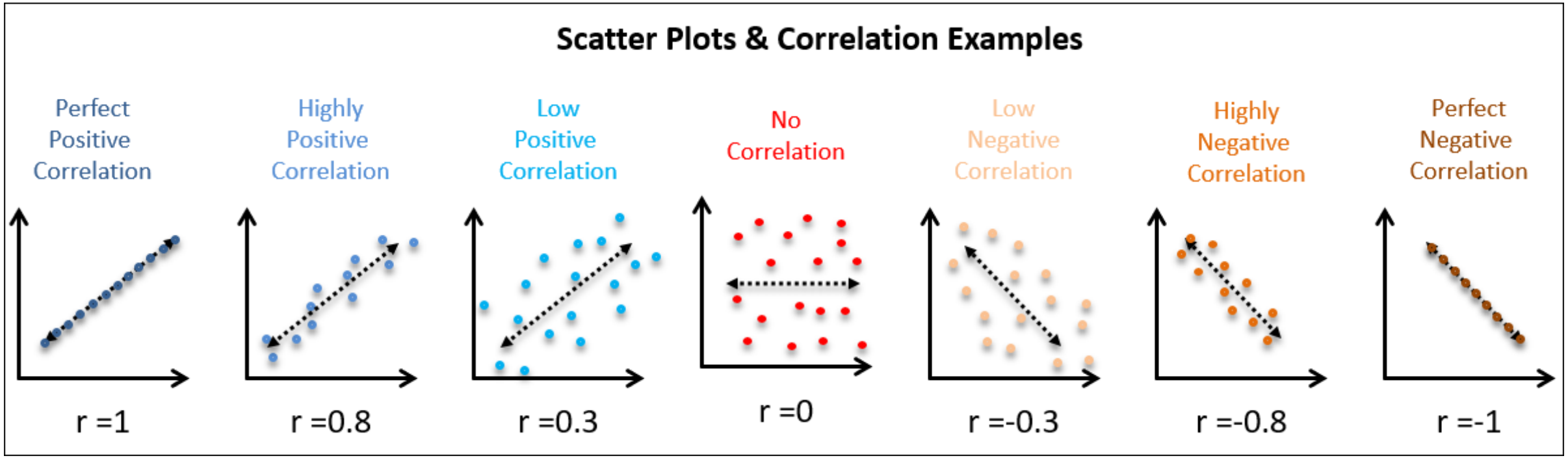
- A scatter plot is a type of data visualization that displays values for two variables as points on a two-dimensional graph.
- Each point represents a pair of values, with one variable plotted on the x-axis and the other on the y-axis



I. Type of chart

Scatter plot: Relationship between two numeric variables

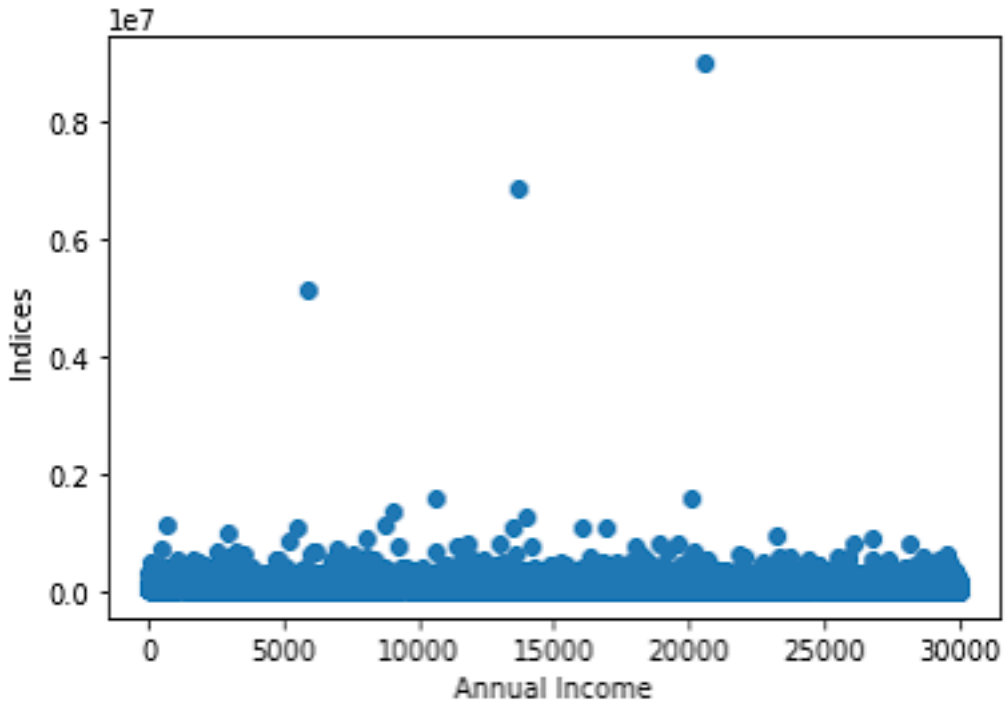
- **Identifying Correlations:** Scatter plots are commonly used to identify correlations between two variables, such as a positive, negative, or no correlation.

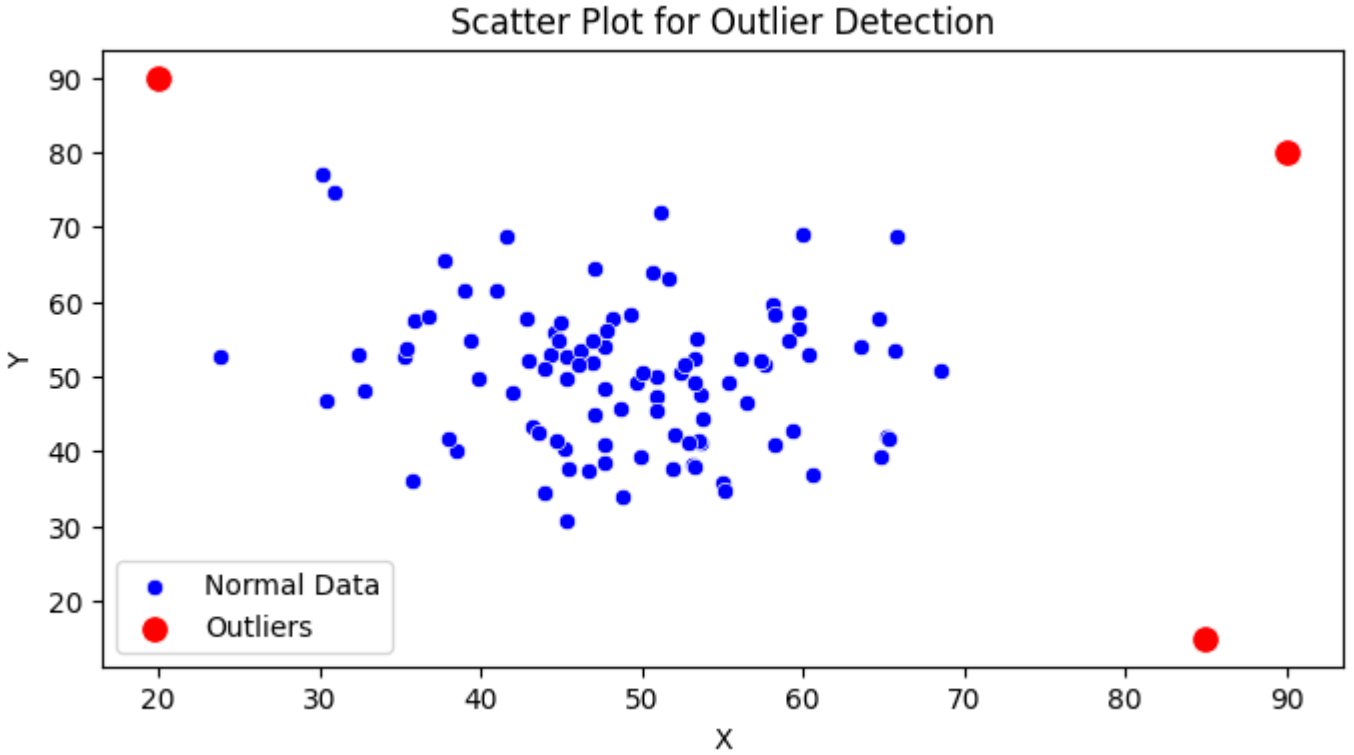


I. Type of chart

Scatter plot: Relationship between two numeric variables

- **Outlier Detection:** Scatter plots can help identify outliers' data points that deviate significantly from the general trend.

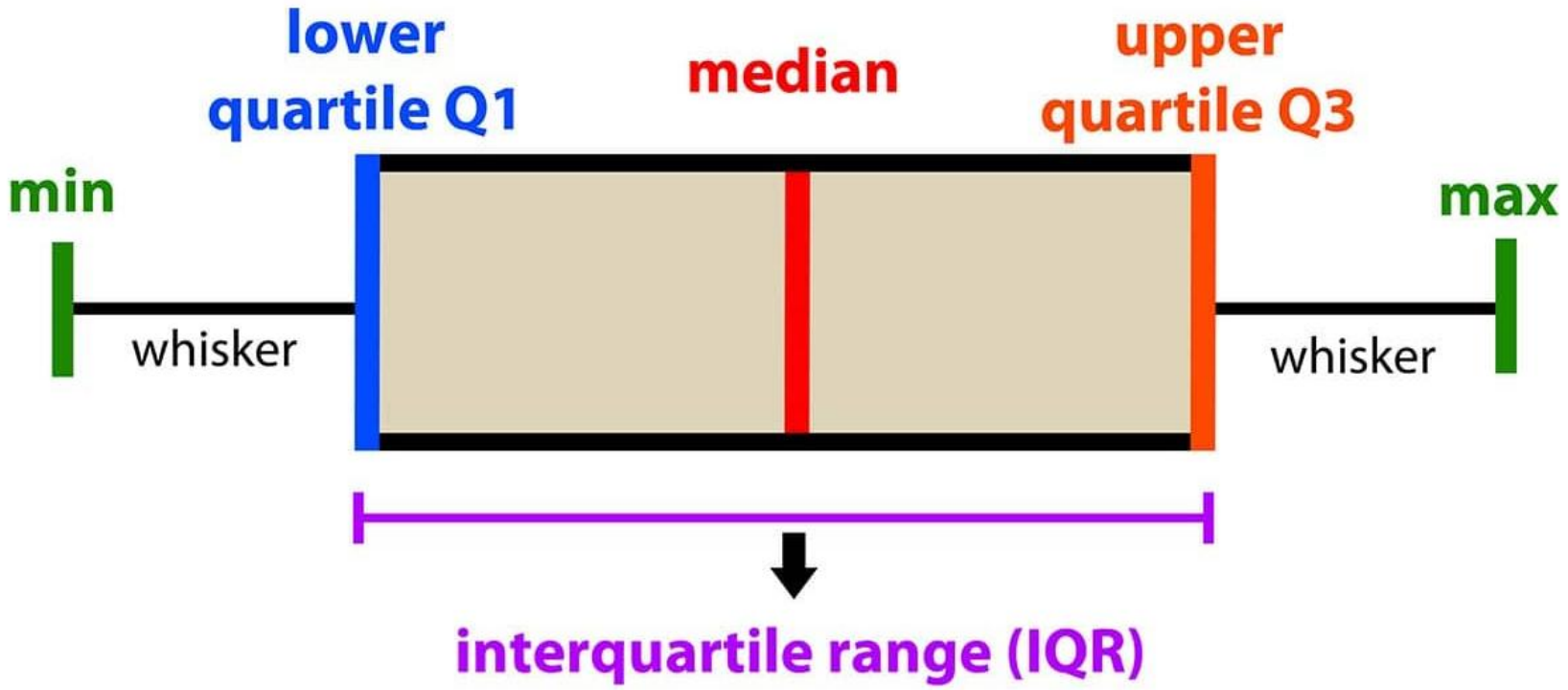




I. Type of chart

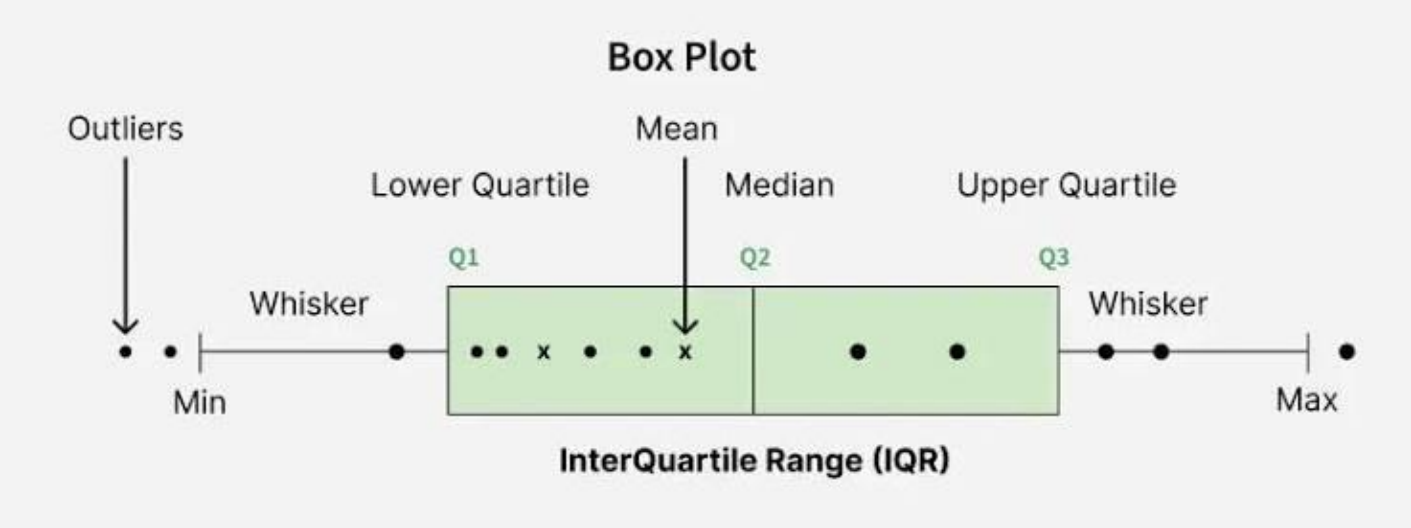
Box plot: Detect outliers, understand spread

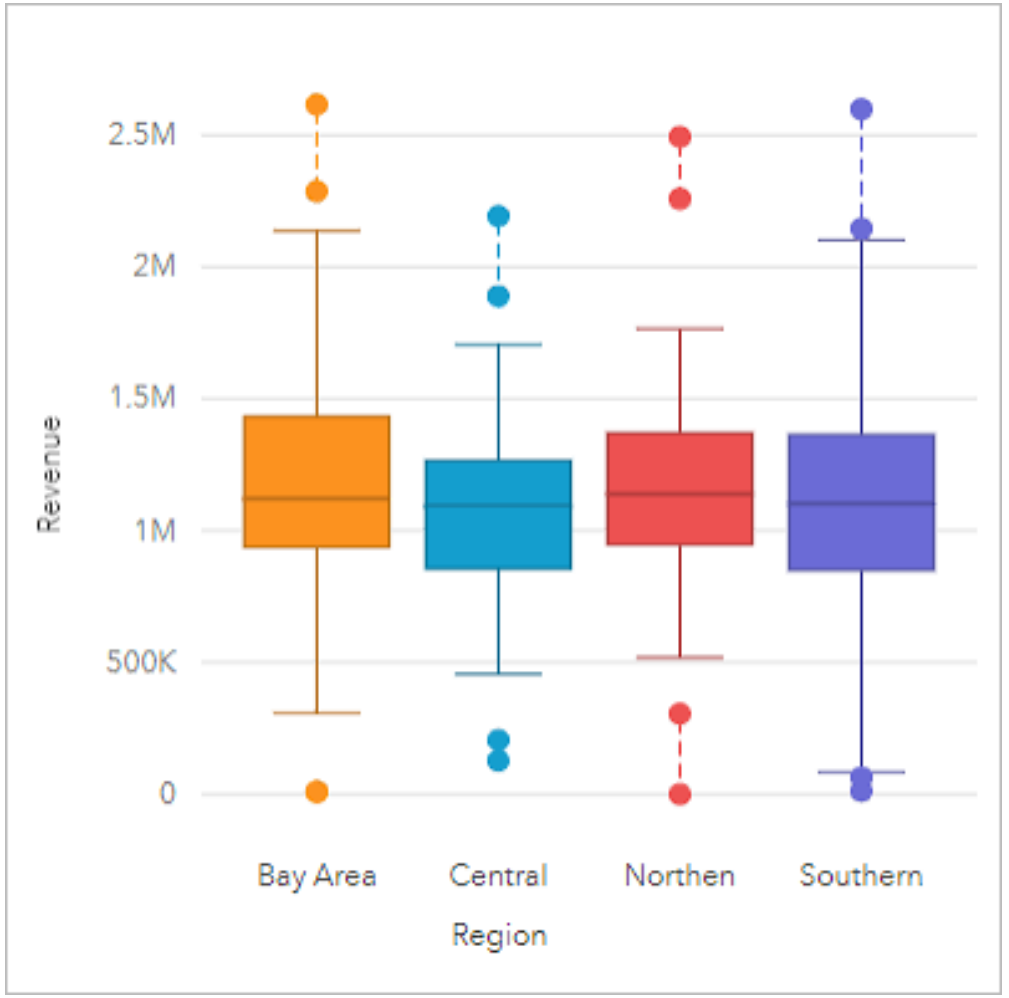
- A box plot (also called a box-and-whisker plot) is a graphical way to show the distribution of a dataset, helping to understand the **spread and central tendency** of the data.



I. Type of chart

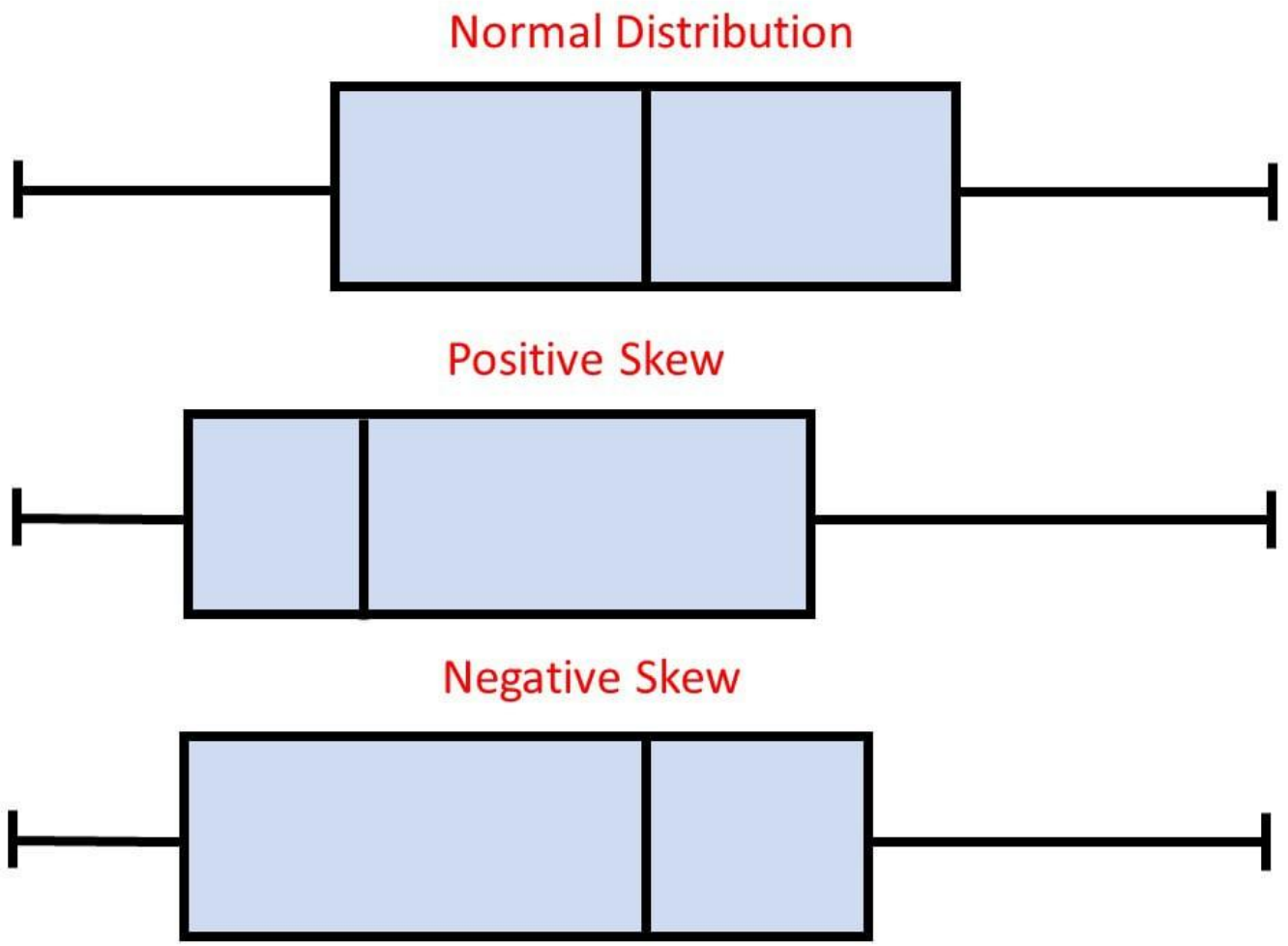
Box plot: Detect outliers





I. Type of chart

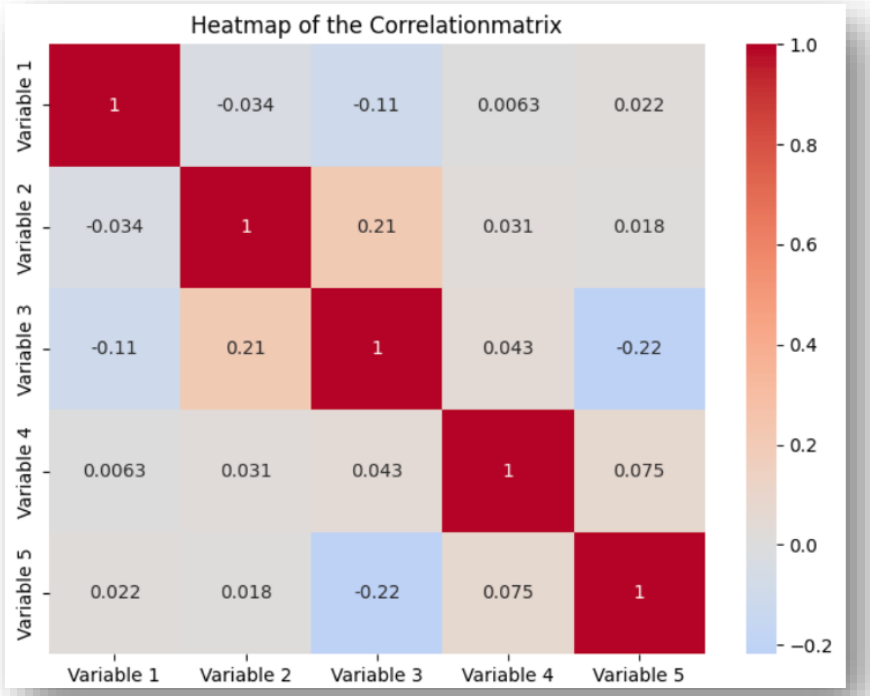
Box plot: Understand spread and distribution

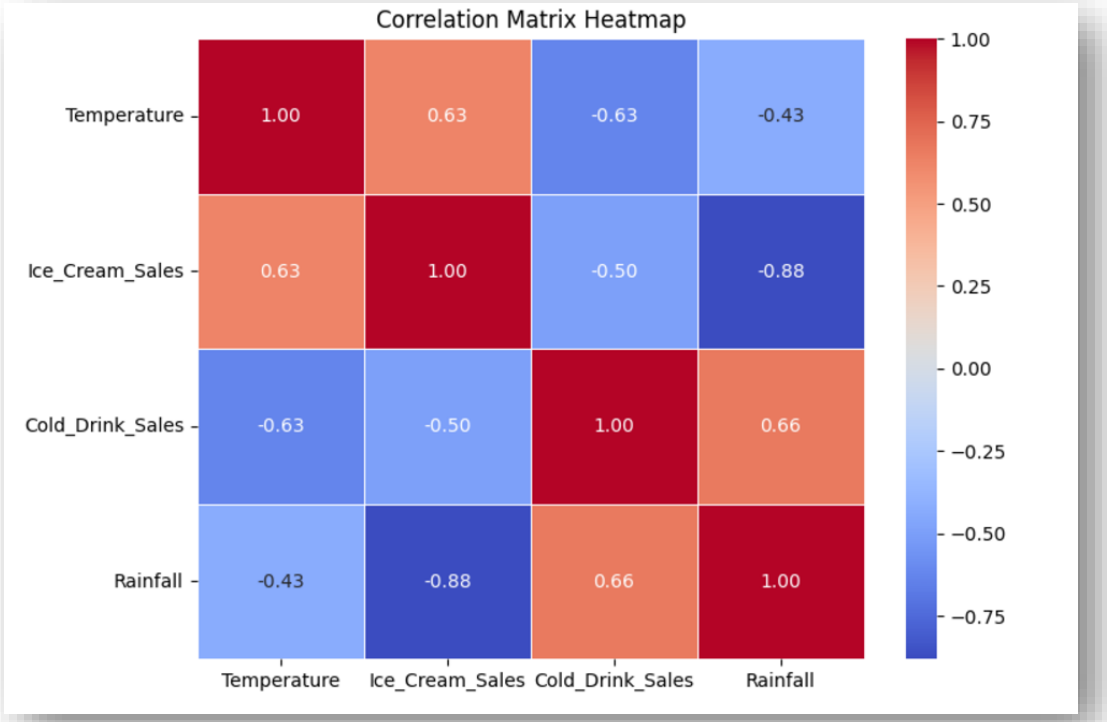


I. Type of chart

Heatmap: Correlation or matrix data visualization

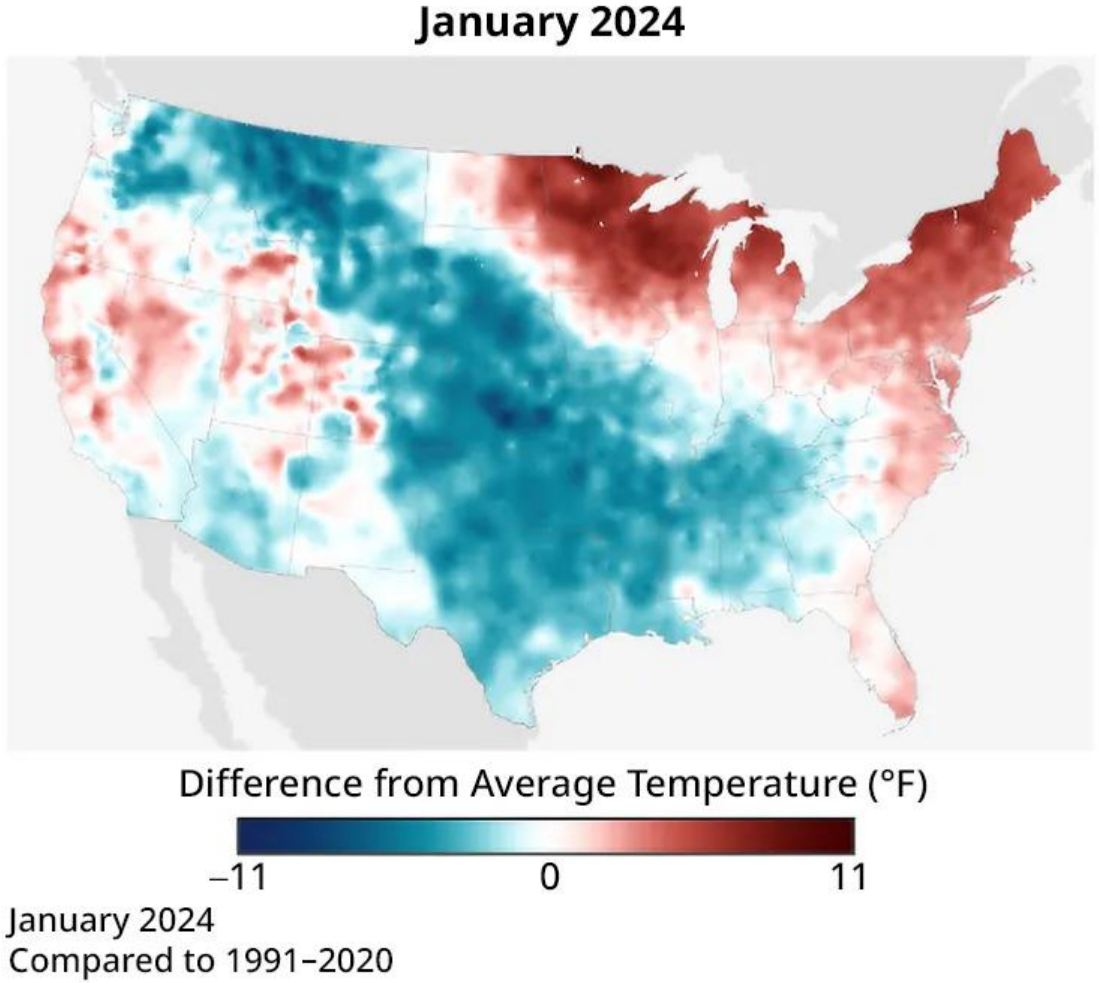
- A heatmap is **a two-dimensional** visualization of data that uses color to represent numerical values. It helps identify relationships between variables.
- Can highlight **highs and lows** in datasets, making trends and patterns easy to spot.





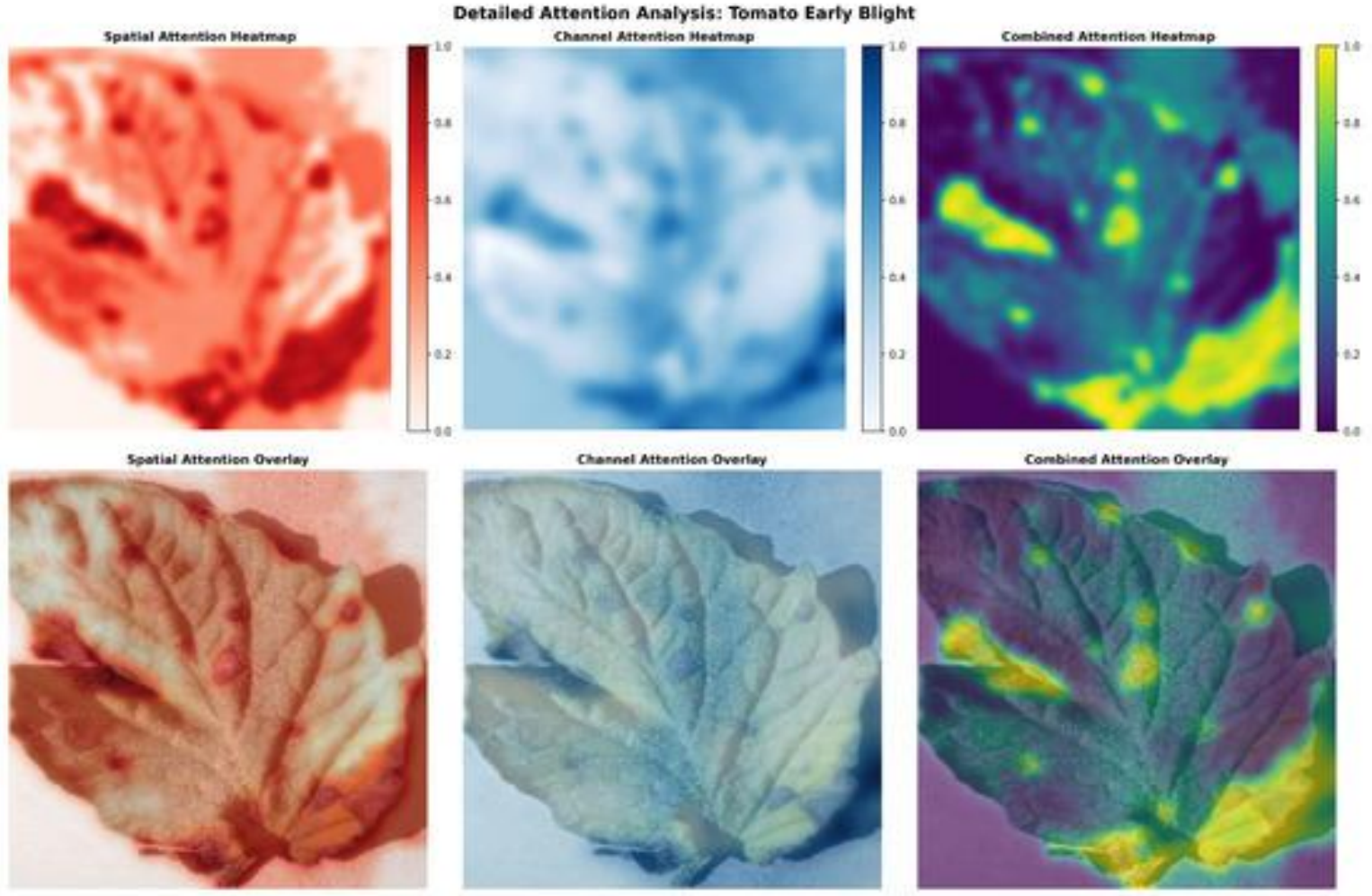
I. Type of chart

Geospatial data in **Heatmap** refers to data that describes the geographic location, shape, size, and other attributes relative to a location on the Earth's surface.



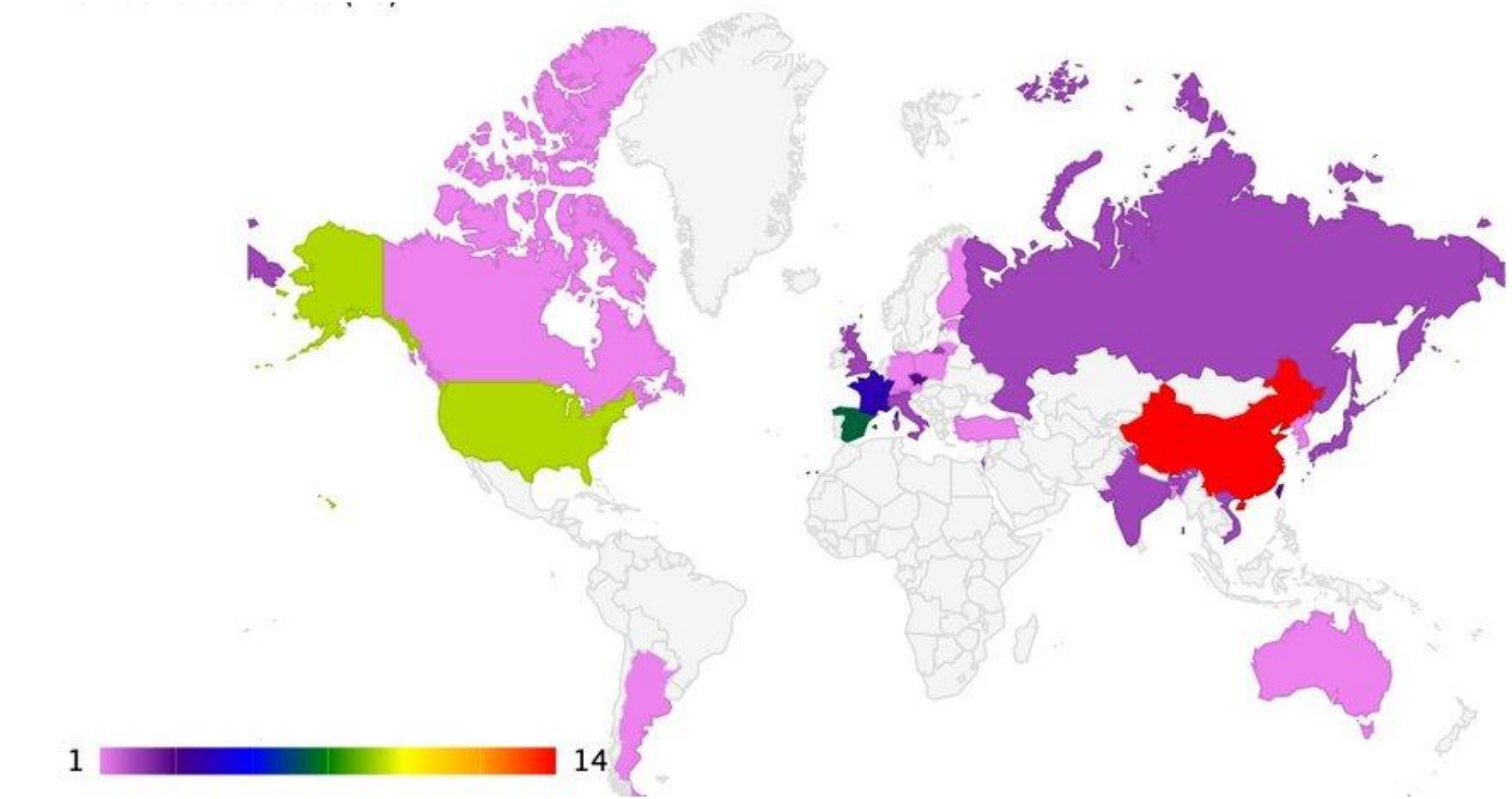
I. Type of chart

Spatial Heatmap: A **spatial heatmap** uses colors to show the **intensity or concentration of something across an area**.



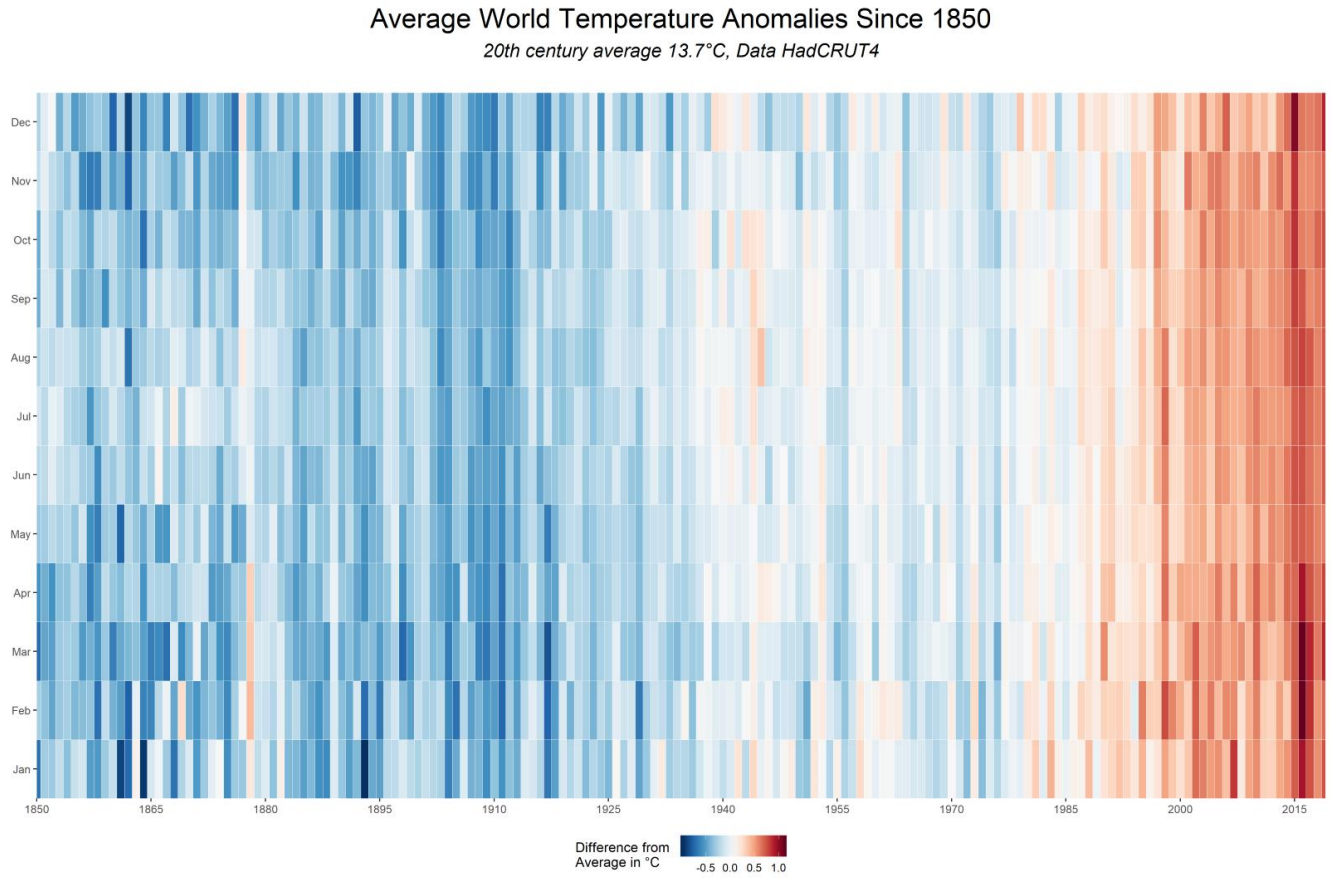
I. Type of chart

Geographic heatmap



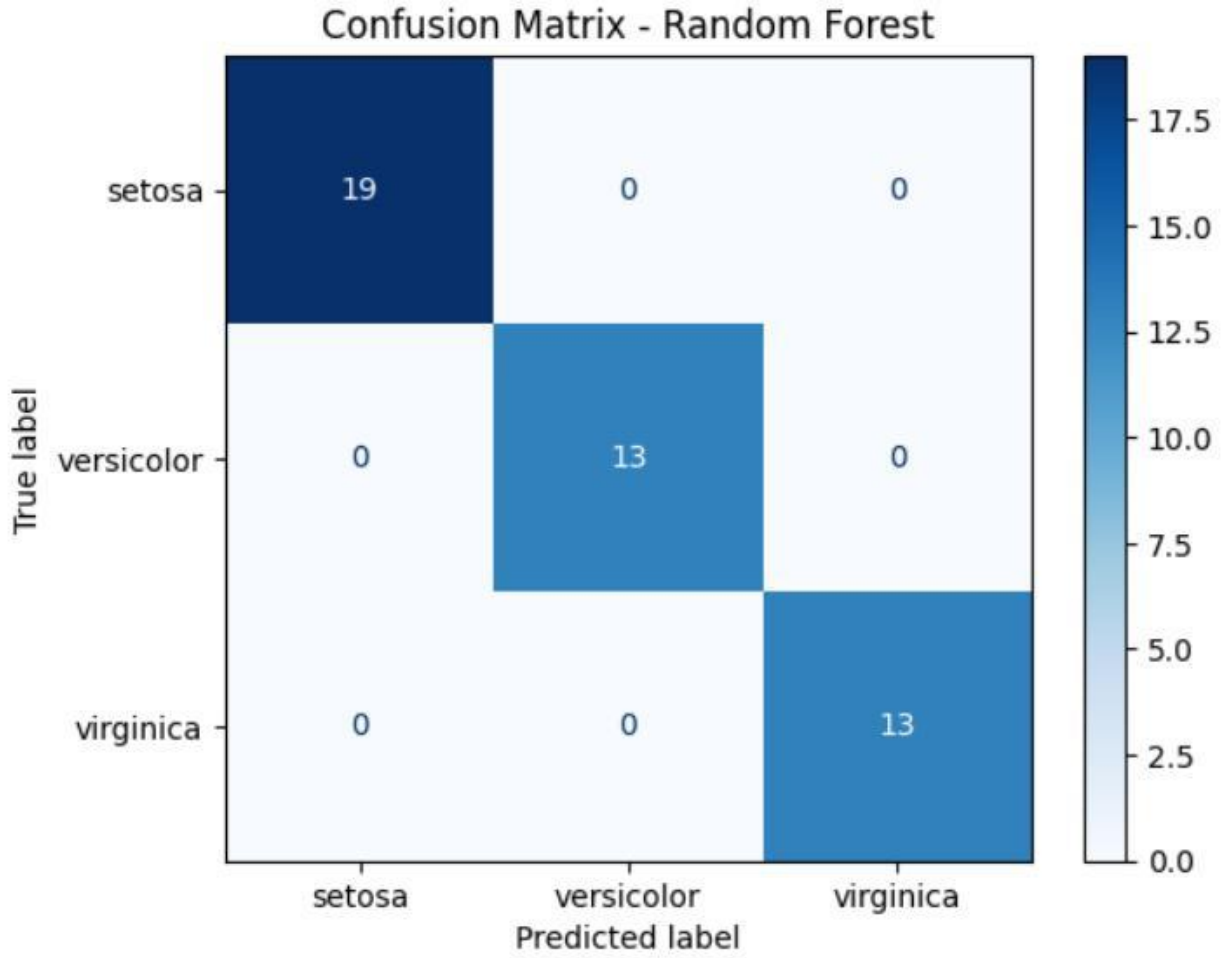
I. Type of chart

Time series Heatmap: Heatmaps can effectively visualize changes over time, and provide an eye-catching alternative to the line chart and heatmaps give us an overview of the broad patterns in the data.



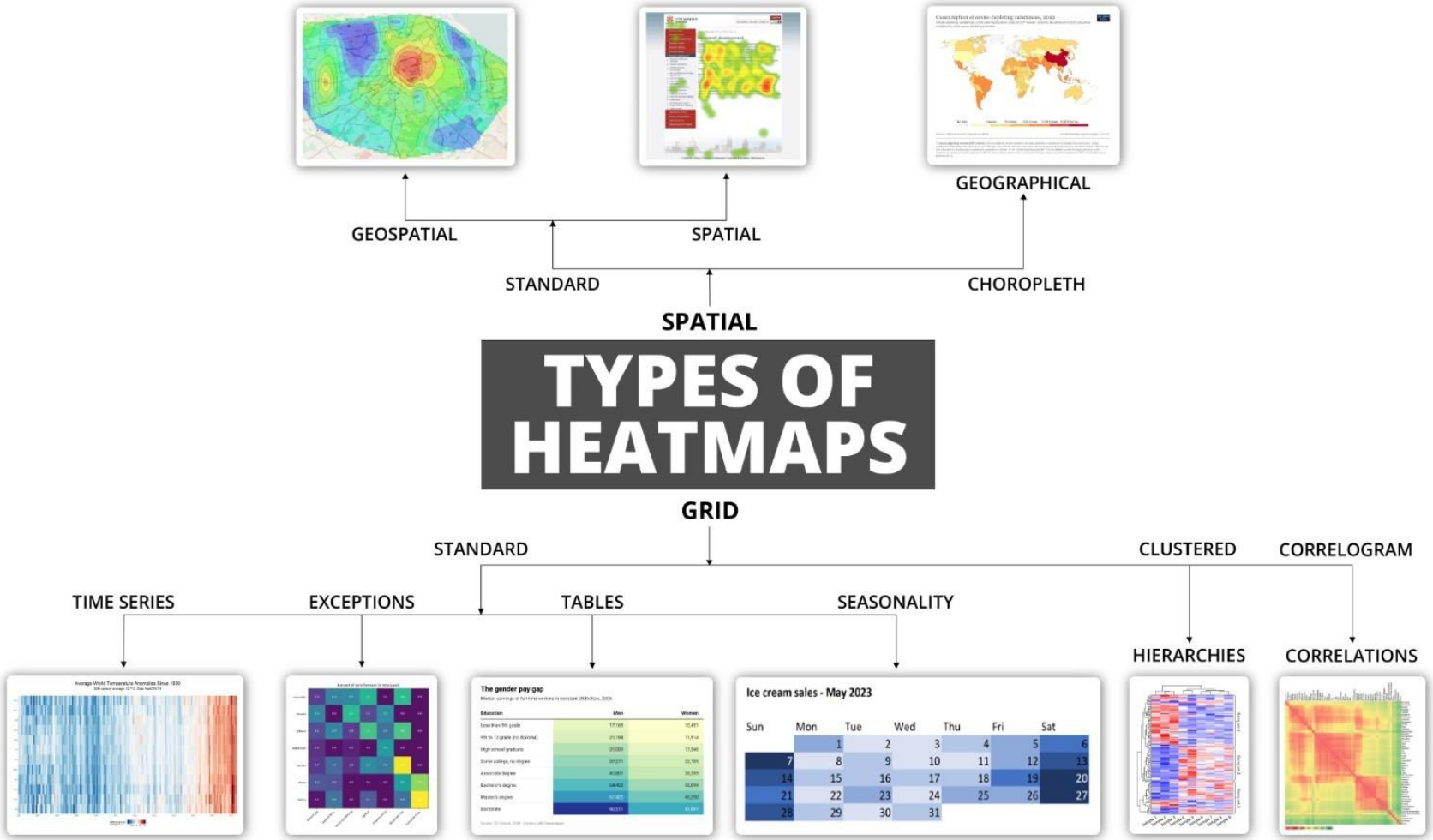
I. Type of chart

Exception Heatmap: it usually refers to a heatmap showing values compared with an **expected value**.



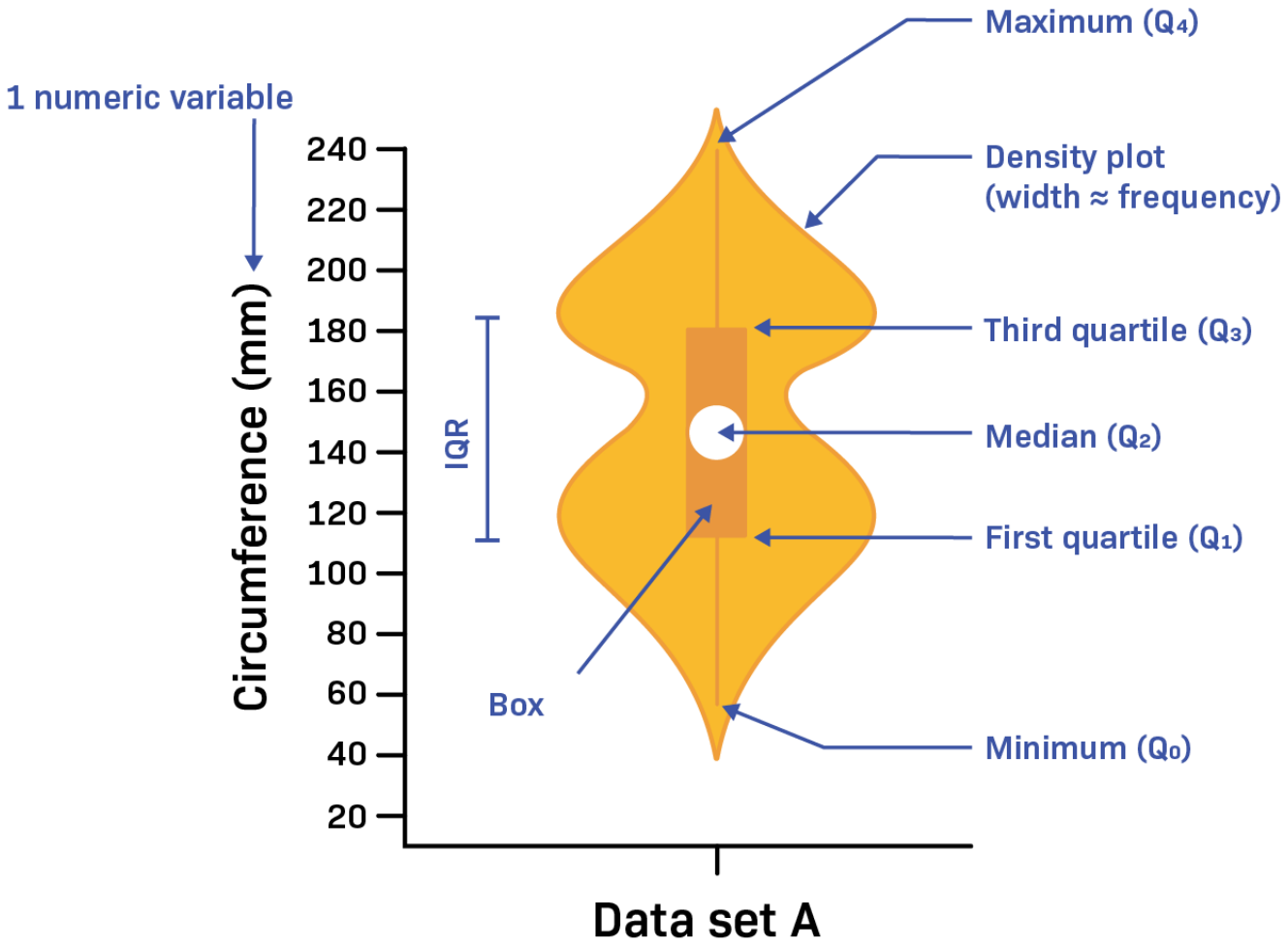
I. Type of chart

Type of Heatmap



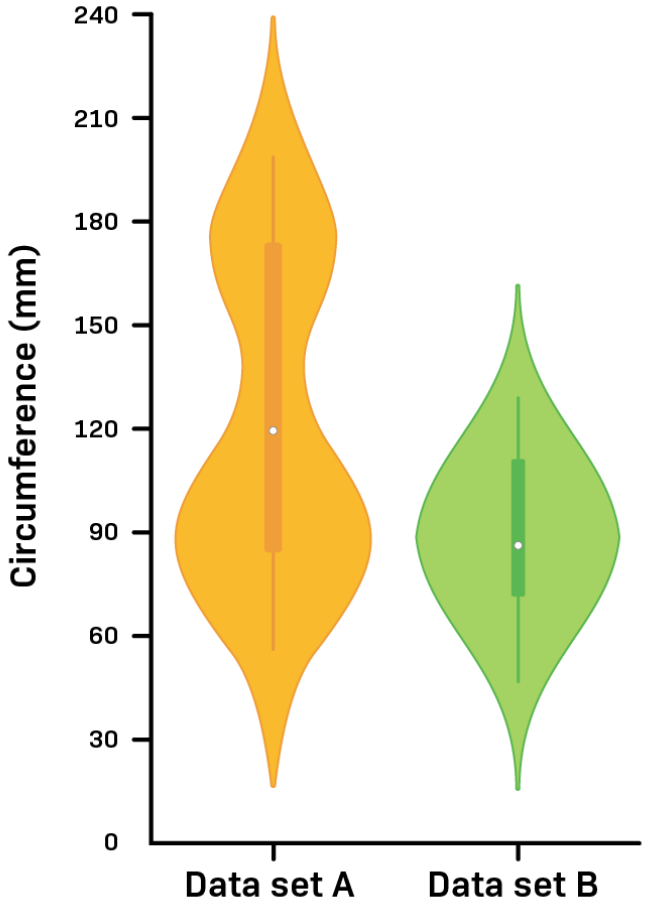
I. Type of chart

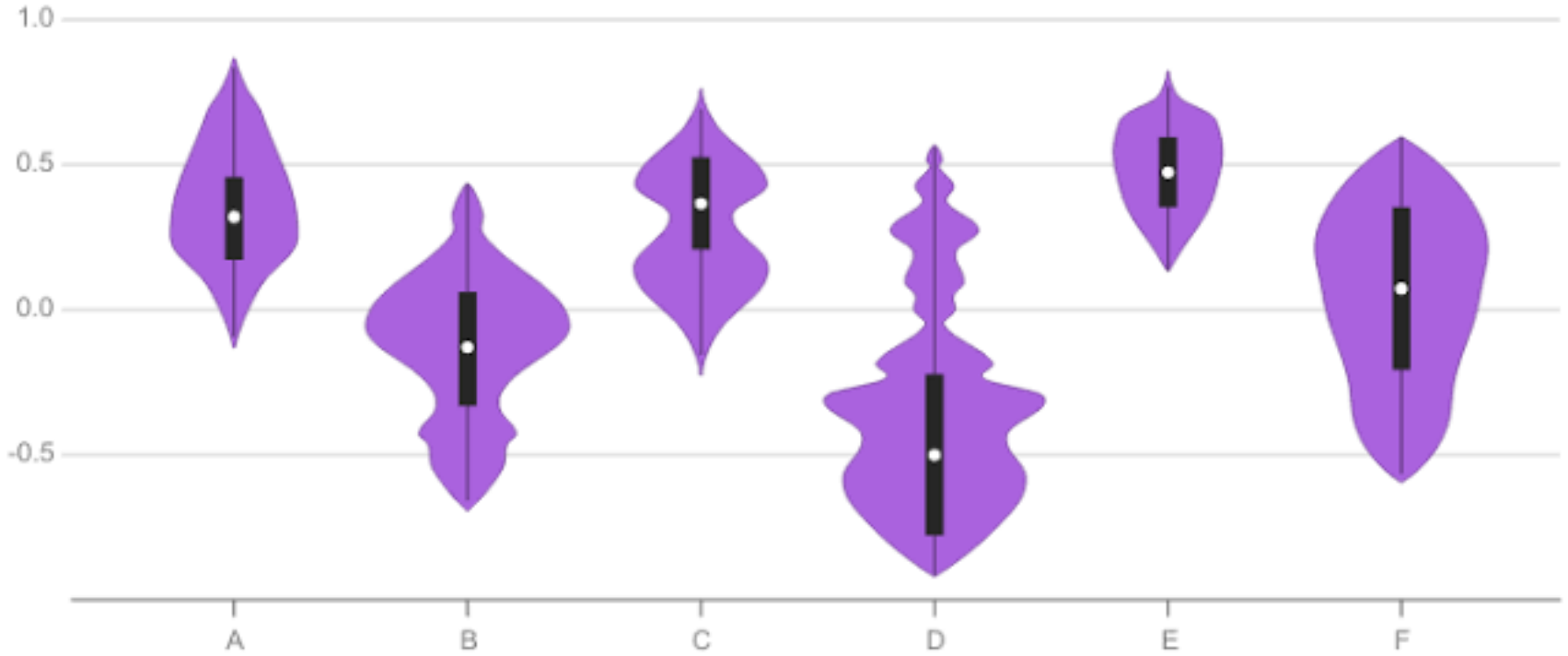
Violin plot: is a method to visualize the distribution of numerical data of different variables.



I. Type of chart

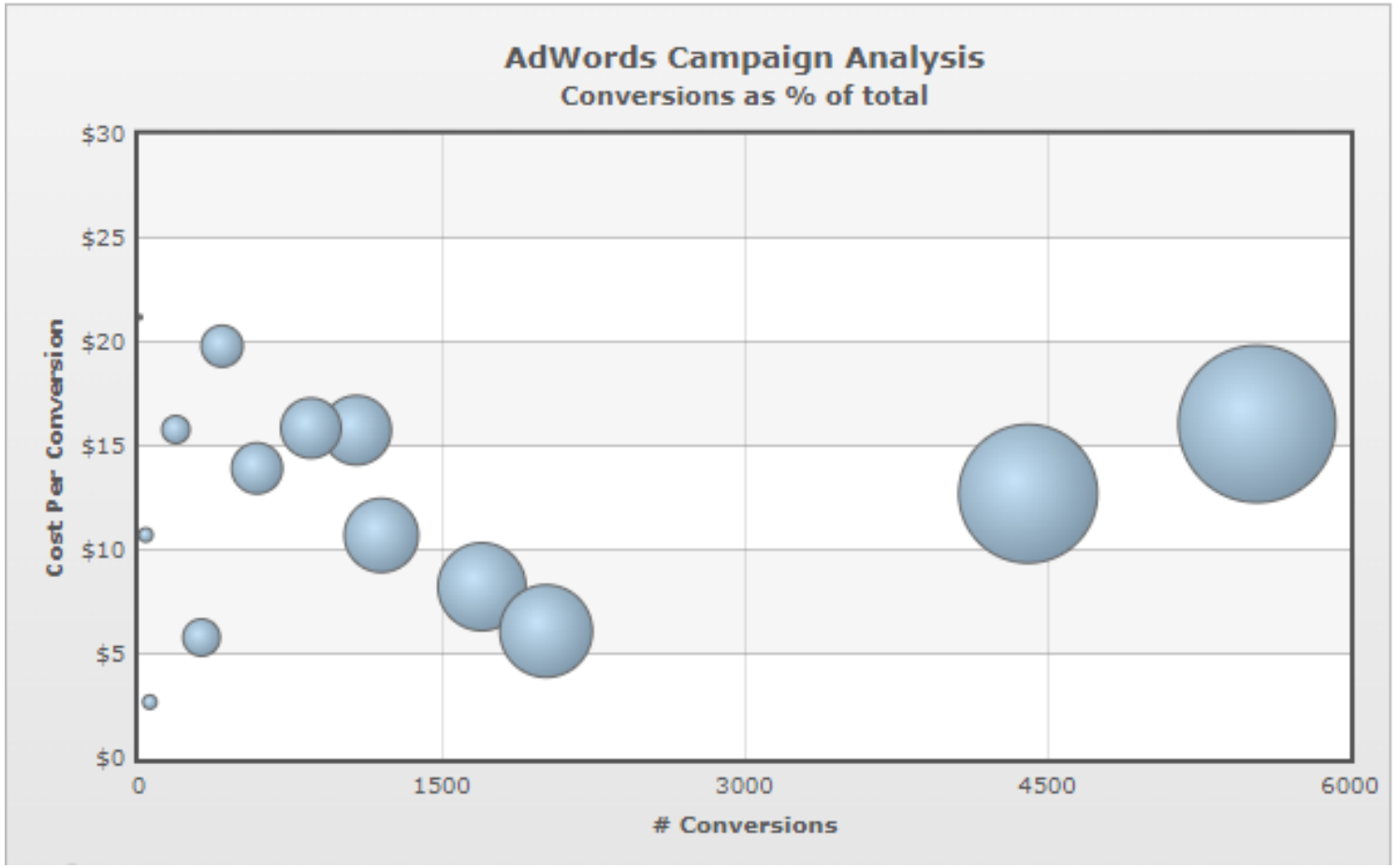
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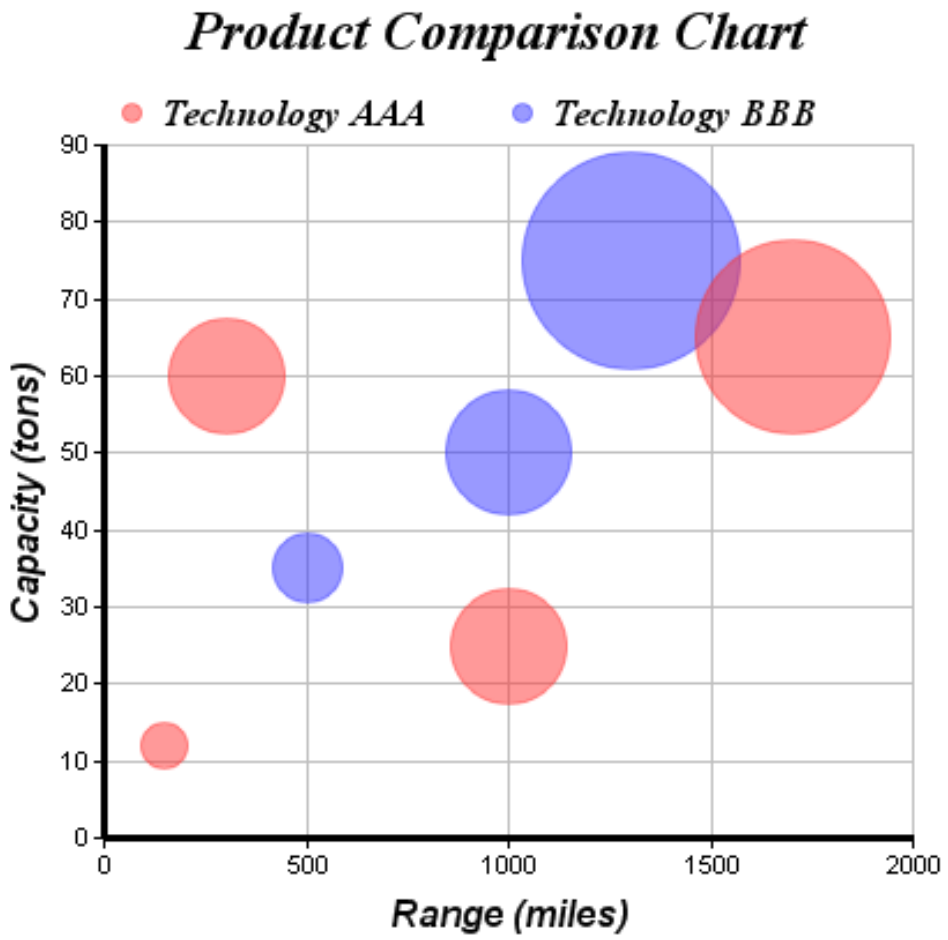
I. Type of chart

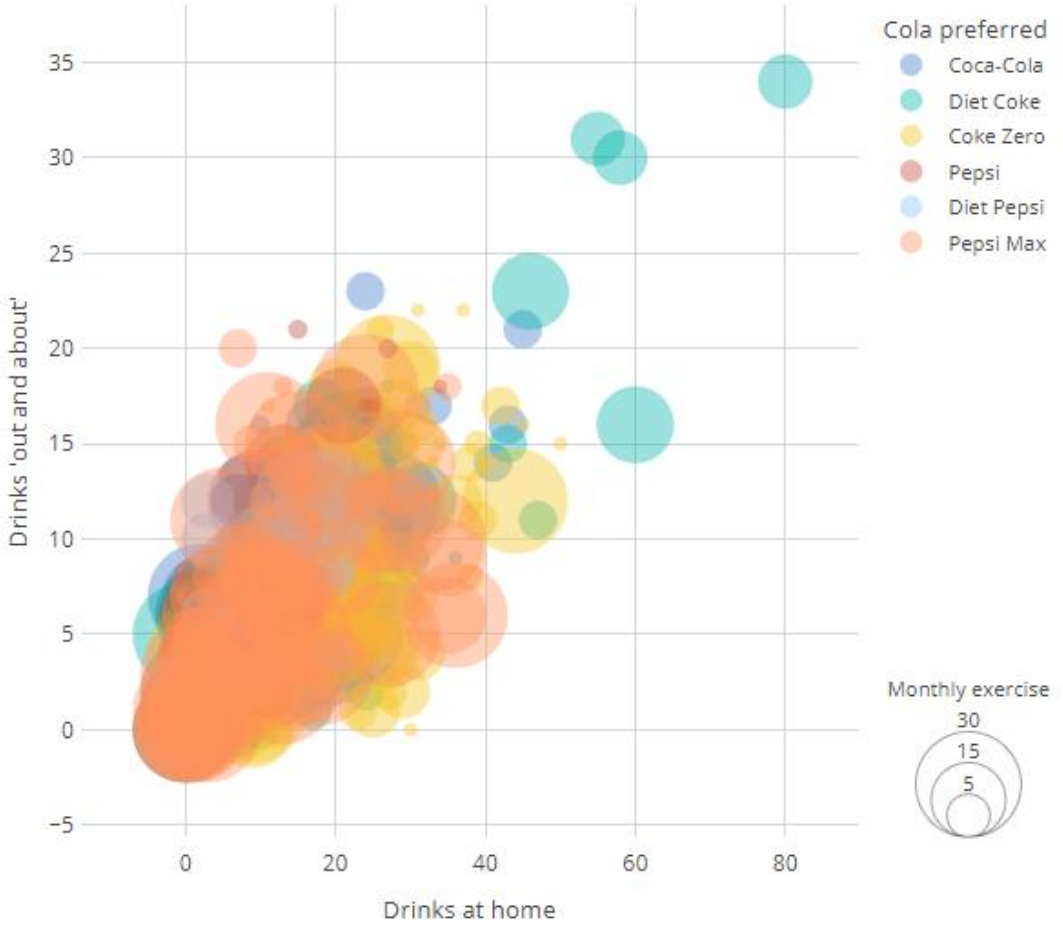
Bubble Chat: is an extension of the scatter plot used to look at relationships between three numeric variables.



I. Type of chart

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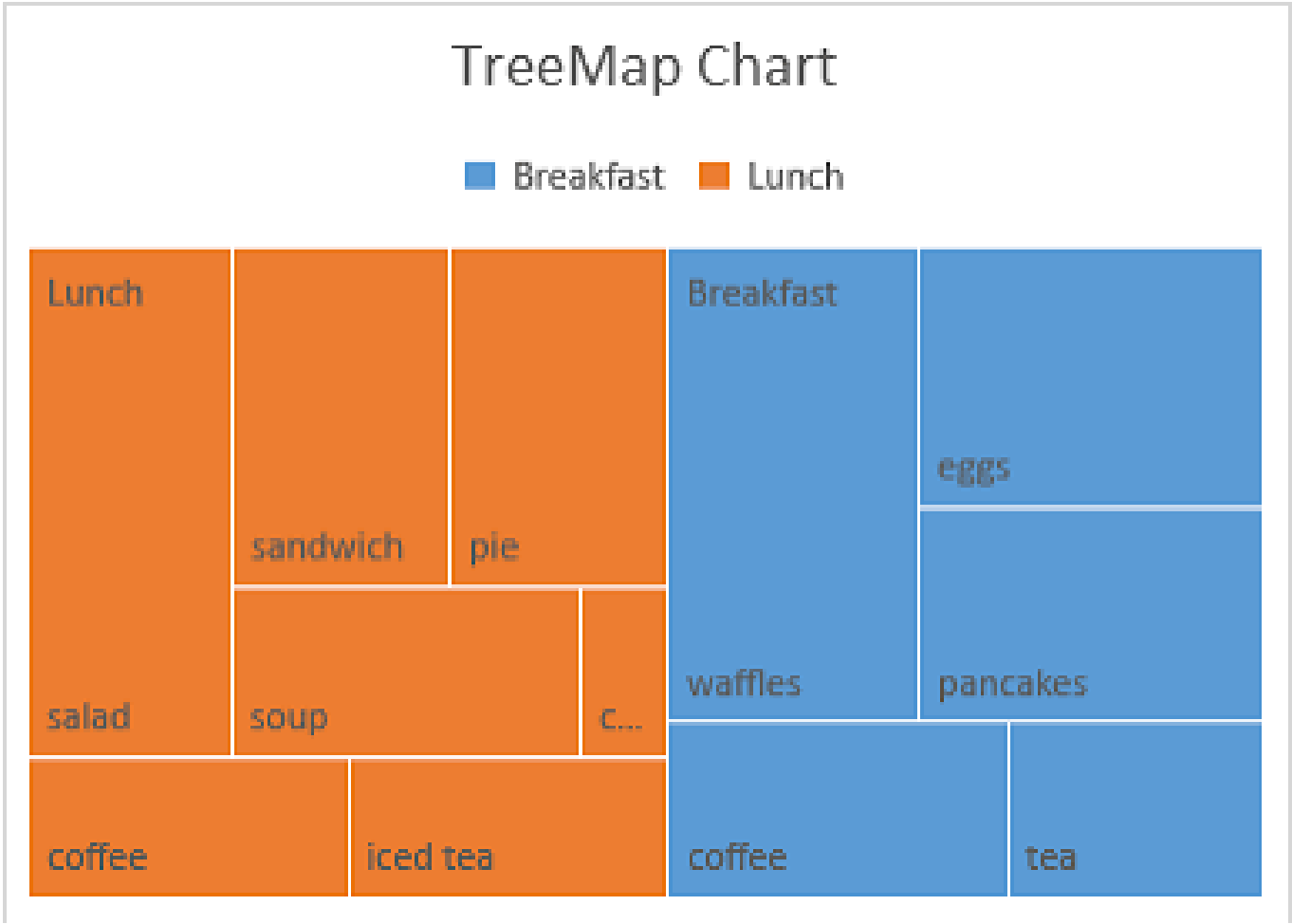
I. Type of chart

Word Clouds Chat: visually displays word frequency and importance in a text body, where the size and prominence of each word indicate how often it appears.



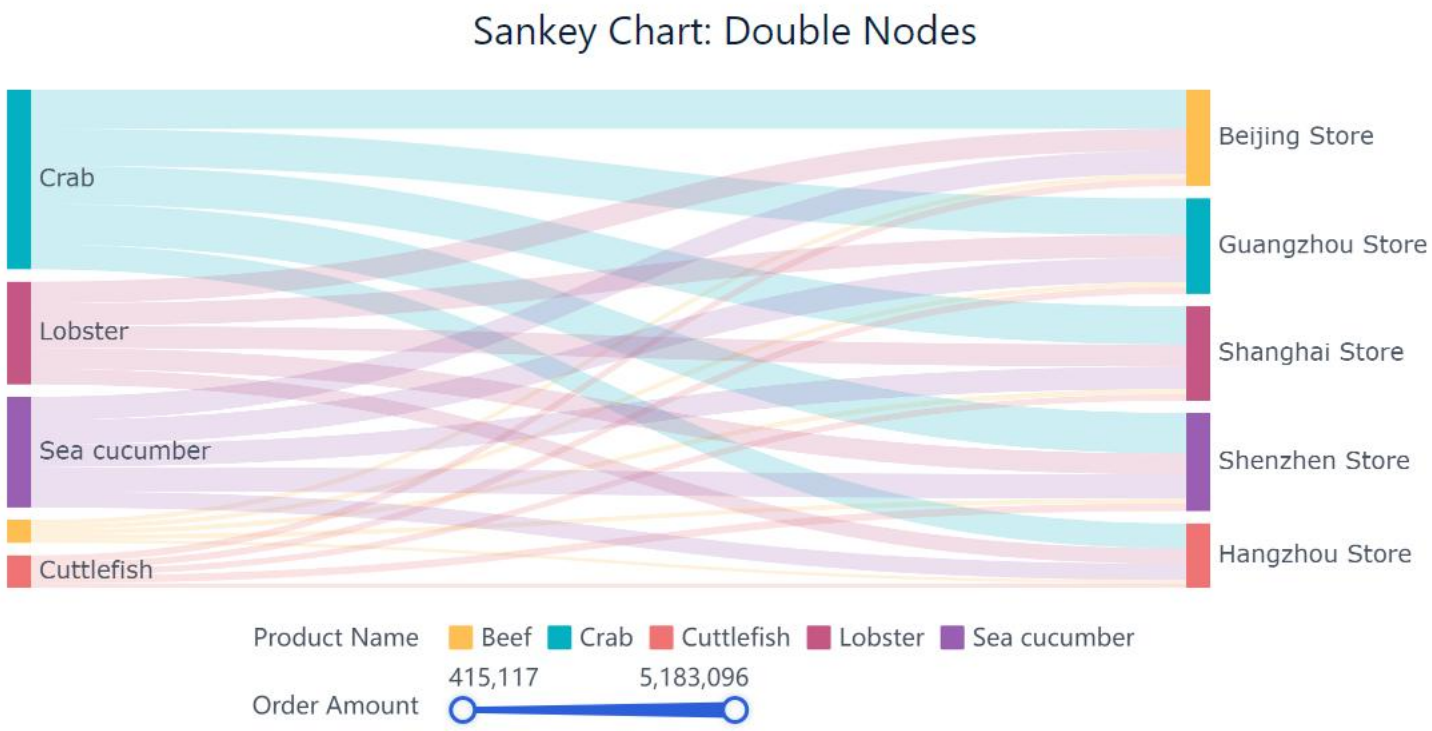
I. Type of chart

Treemap Chart: uses nested rectangles to display hierarchical data and part-to-whole proportions, where each box's size represents a quantitative value.



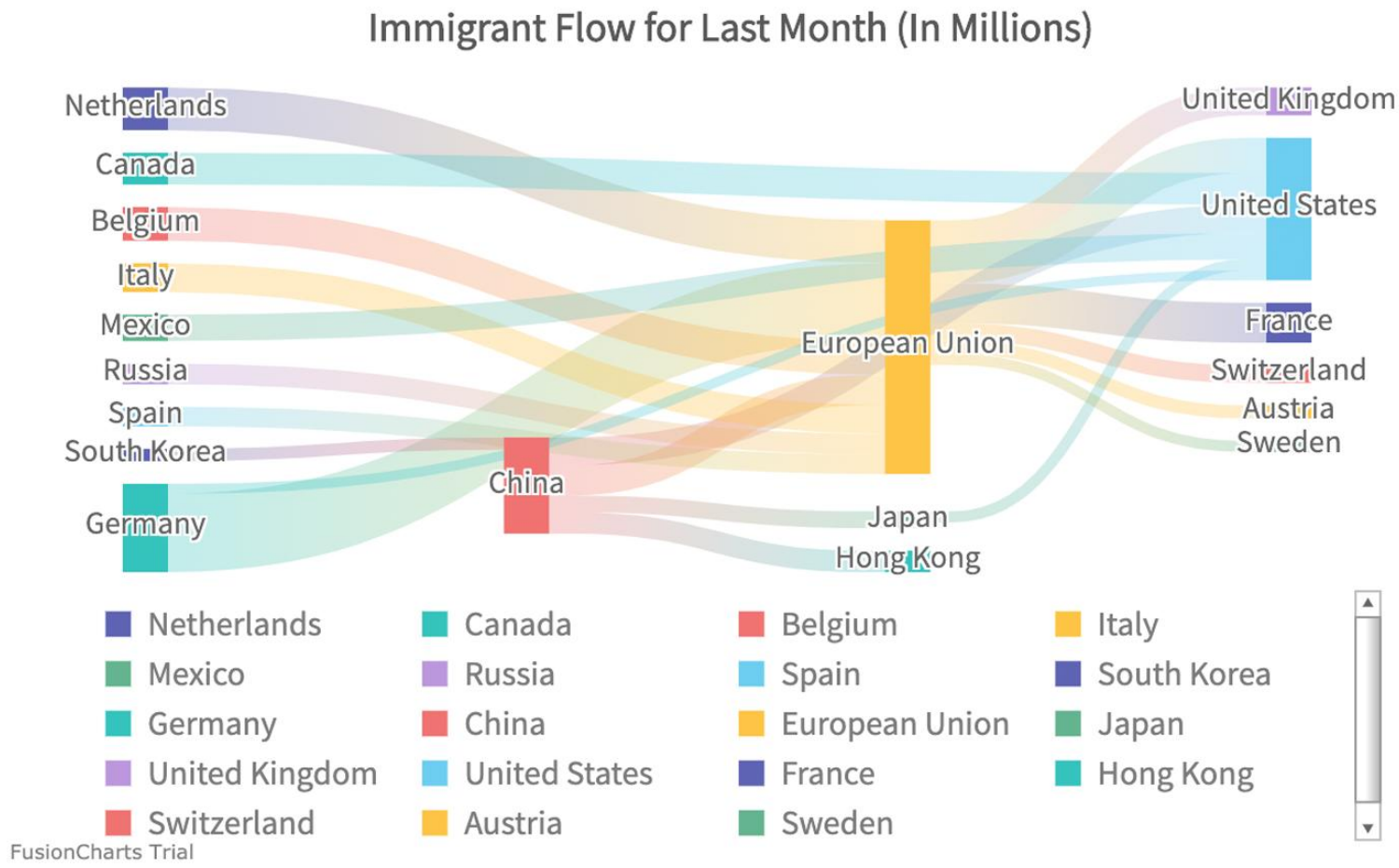
I. Type of chart

Sankey Chart: shows flows between entities, with line thickness representing volume.



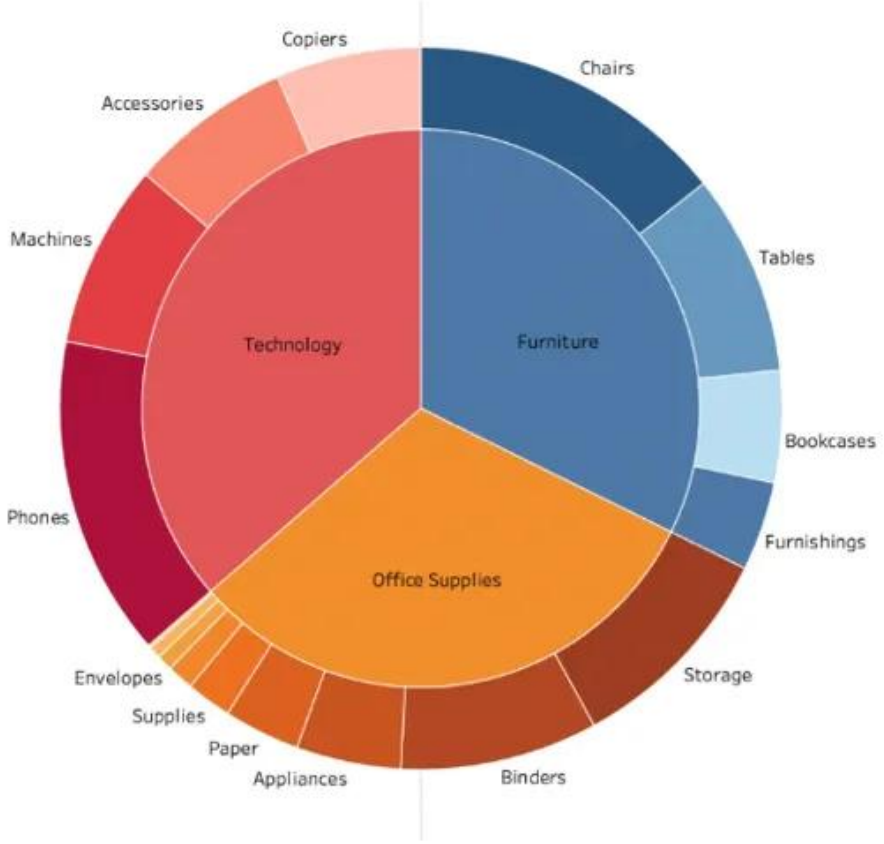
I. Type of chart

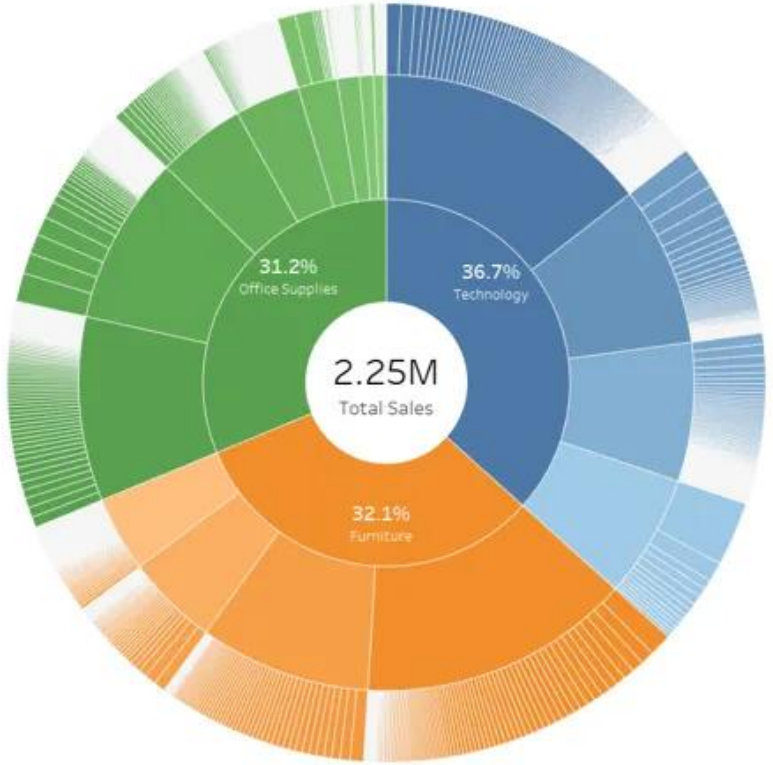
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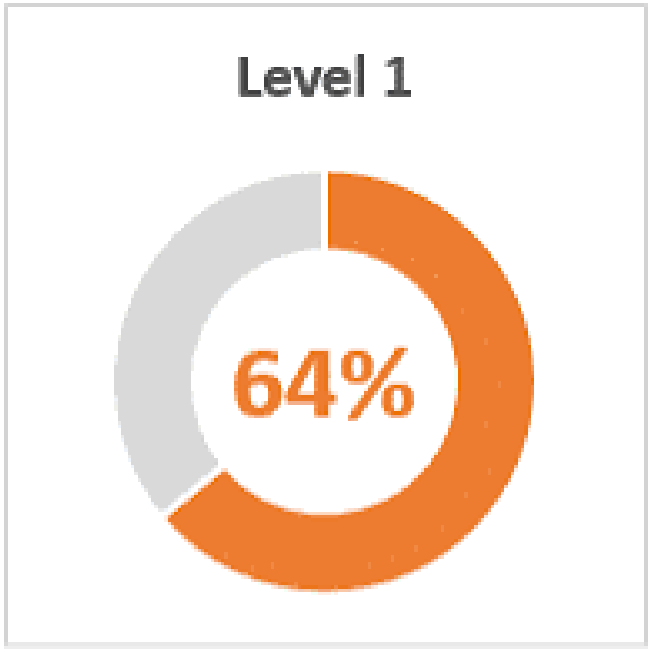
Sunburst Chart: is a visualization used to show **hierarchical data,** data organized into into levels or parent-child relationships.

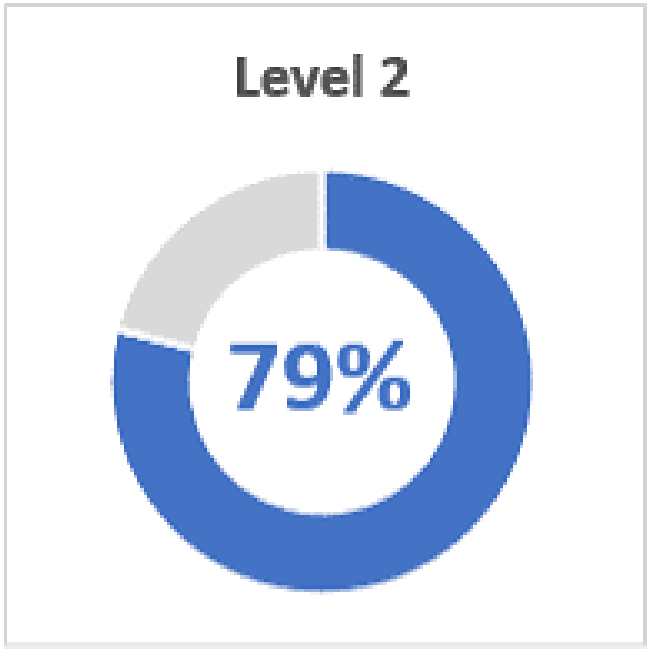


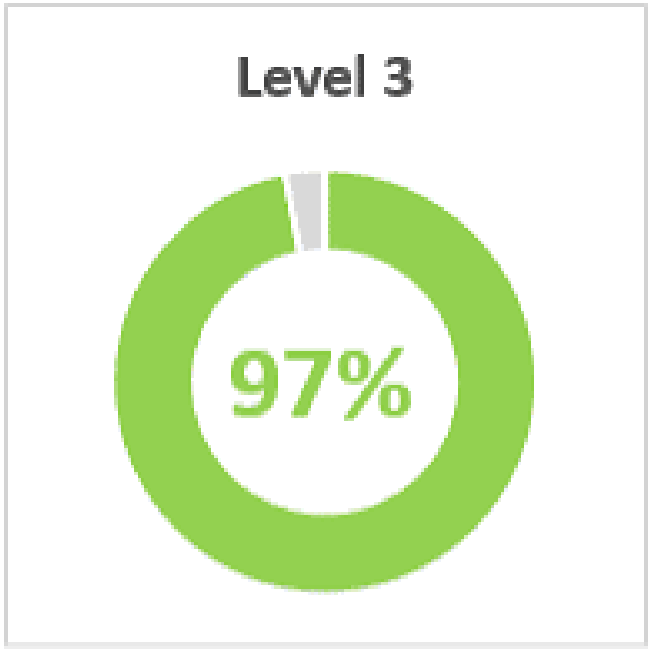


I. Type of chart

Progress Circle Chart: A Progress Circle Chart displays progress toward a goal with a circular indicator.



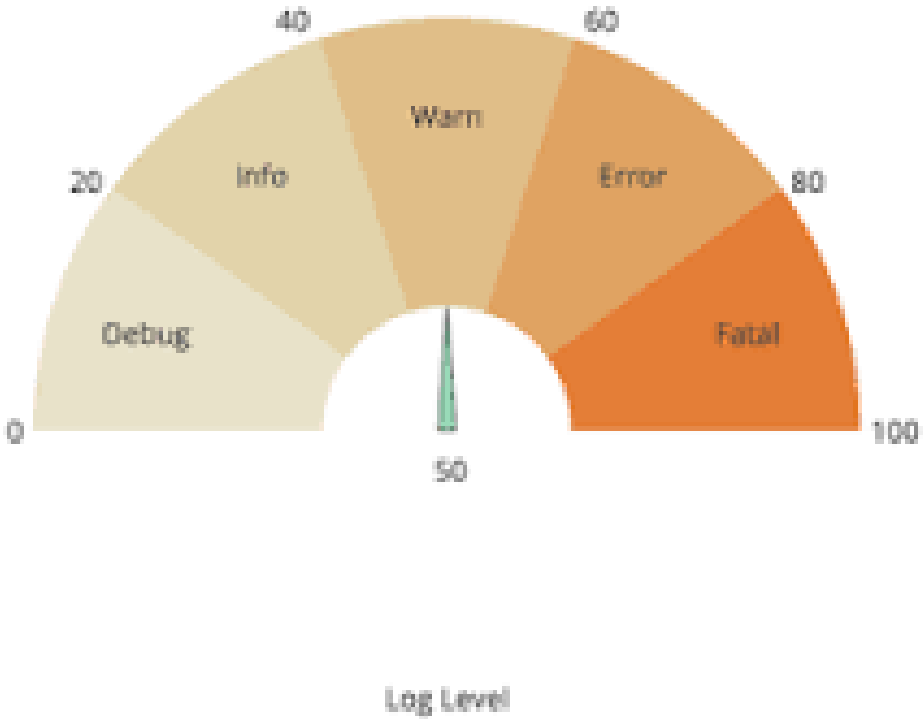




I. Type of chart

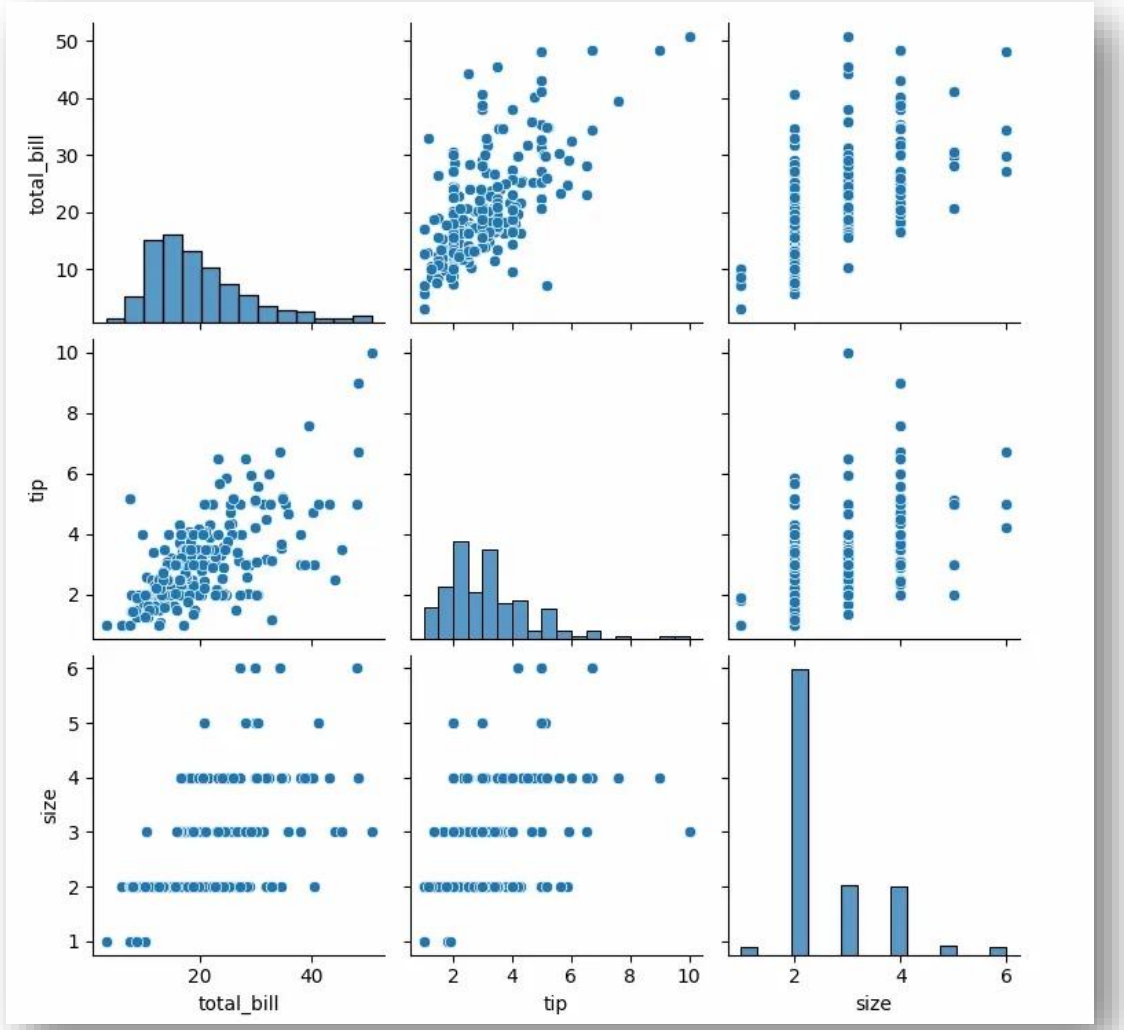
Gauge Chart: is a visualization used to show **one value compared with a target, limit, or scale**.





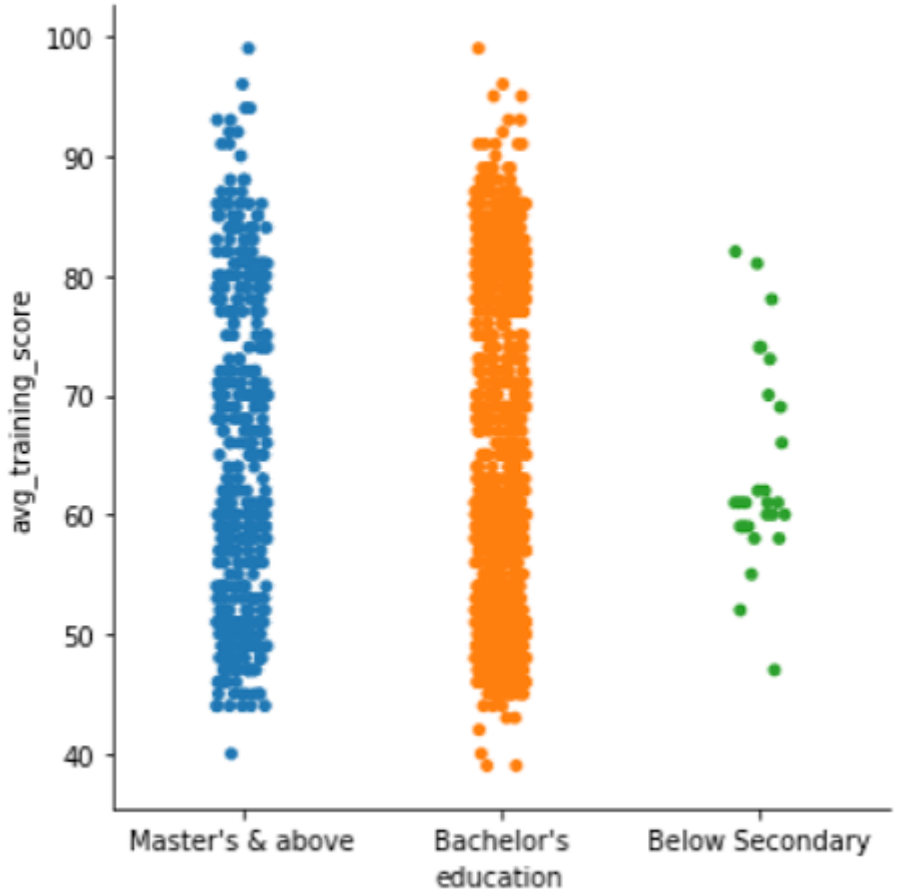
I. Type of chart

Pair plot: Exploring Relationships Between Two Variables.



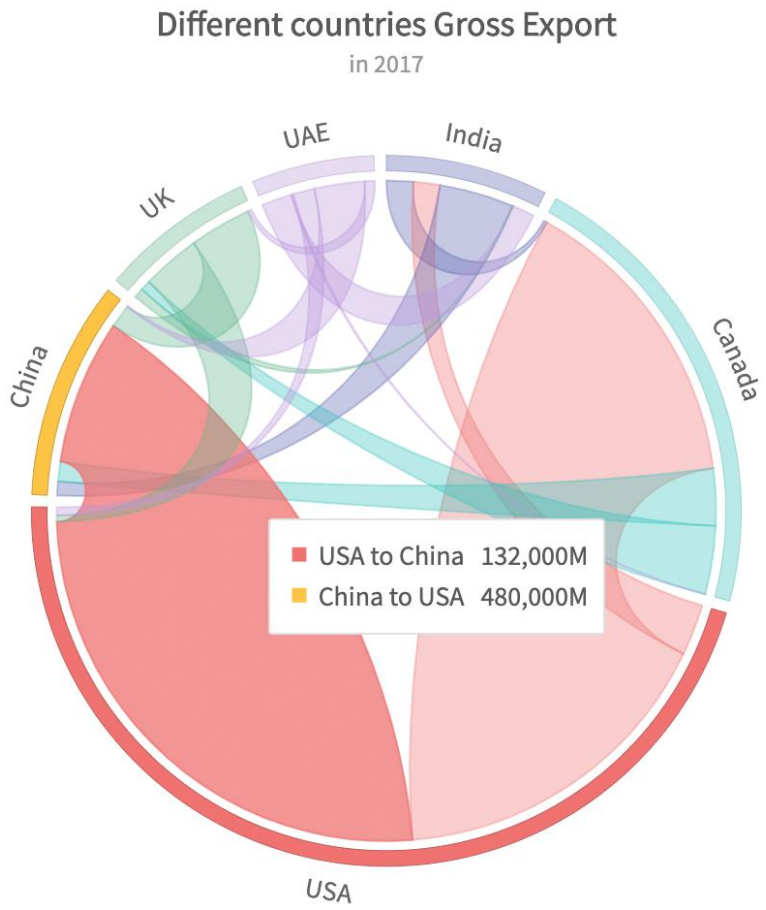
I. Type of chart

catplot: show the relationship between a numerical and one or more categorical variables.



I. Type of chart

Chord diagram: show relationships or connections between categories.



II. Colour Theory

❑ Fundamentals of Colour Theory

Colour theory is the scientific and subjective study of color to understand its effects on human perception and how to use it to improve communication and design decisions.

1. Colour Anatomy

❖ Study of the components and perceptions of colour

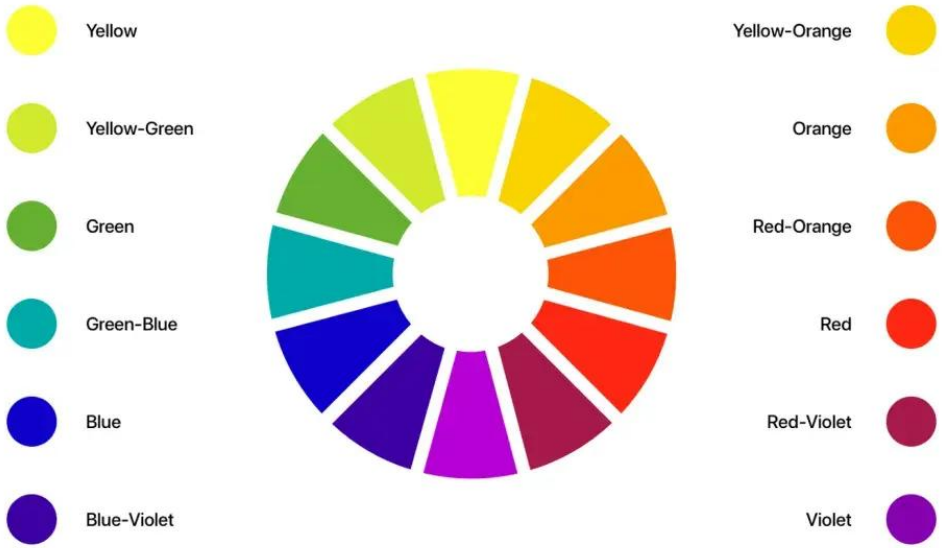
❖ Basic components of colour

❖ There are 3 type of Colour Anatomy is **Hue, Saturation, and lightness**.

II. Colour Theory

❑ Hue

- ❖ Used to distinguish different categories
- ❖ Basic color such as red, blue, green, or yellow





Hue

II. Colour Theory

❑ Saturation

- ❖ Is the ratio of a color's intensity to its area of coverage
- ❖ High saturation level are vivid and eye-catching
- ❖ Use lower saturation for less prominent elements





II. Colour Theory

❑ Lightness

- ❖ Use a light-to-dark scale to represent different numerical values.
- ❖ Lightness describes how light or dark a colour appears.
- ❖ Tints are produced by adding white, while shades are produced by adding black.



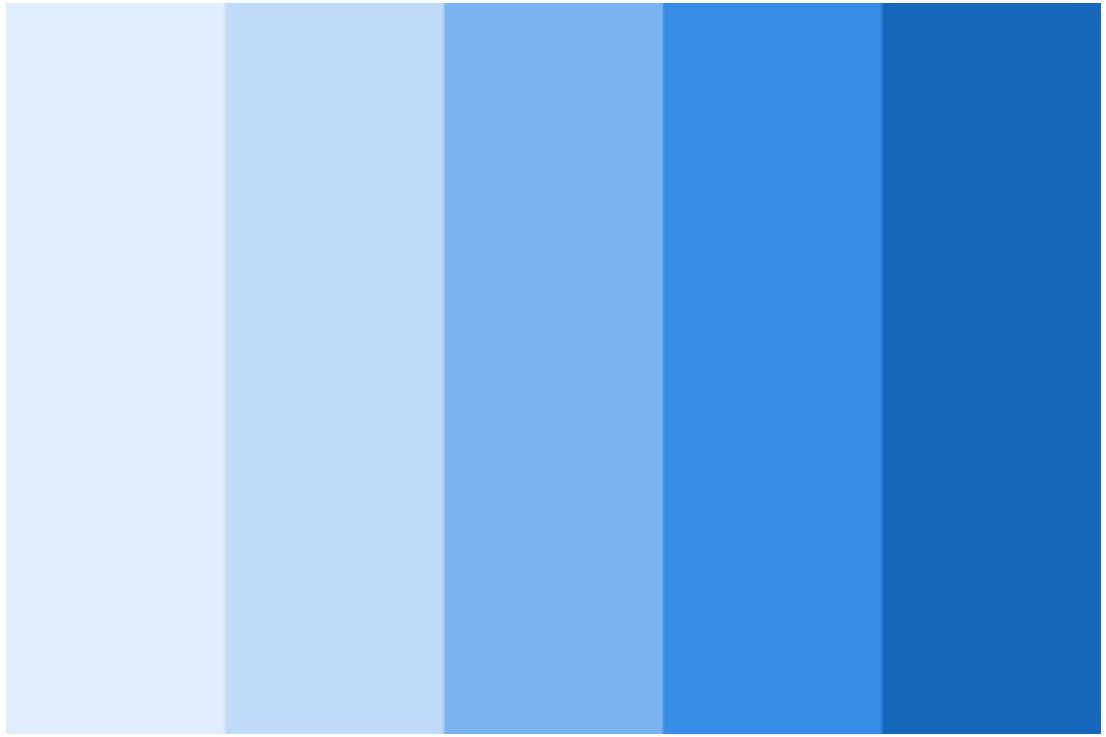
II. Colour Theory

2. Color Harmony

- Specific colour combinations.
- Can provide either visual contrast or cohesiveness.
- There are 6 type of **Color Harmony is** Monochromatic, Analogous, Complementary, Split-complementary, Triadic, and Tetradic.

II. Colour Theory

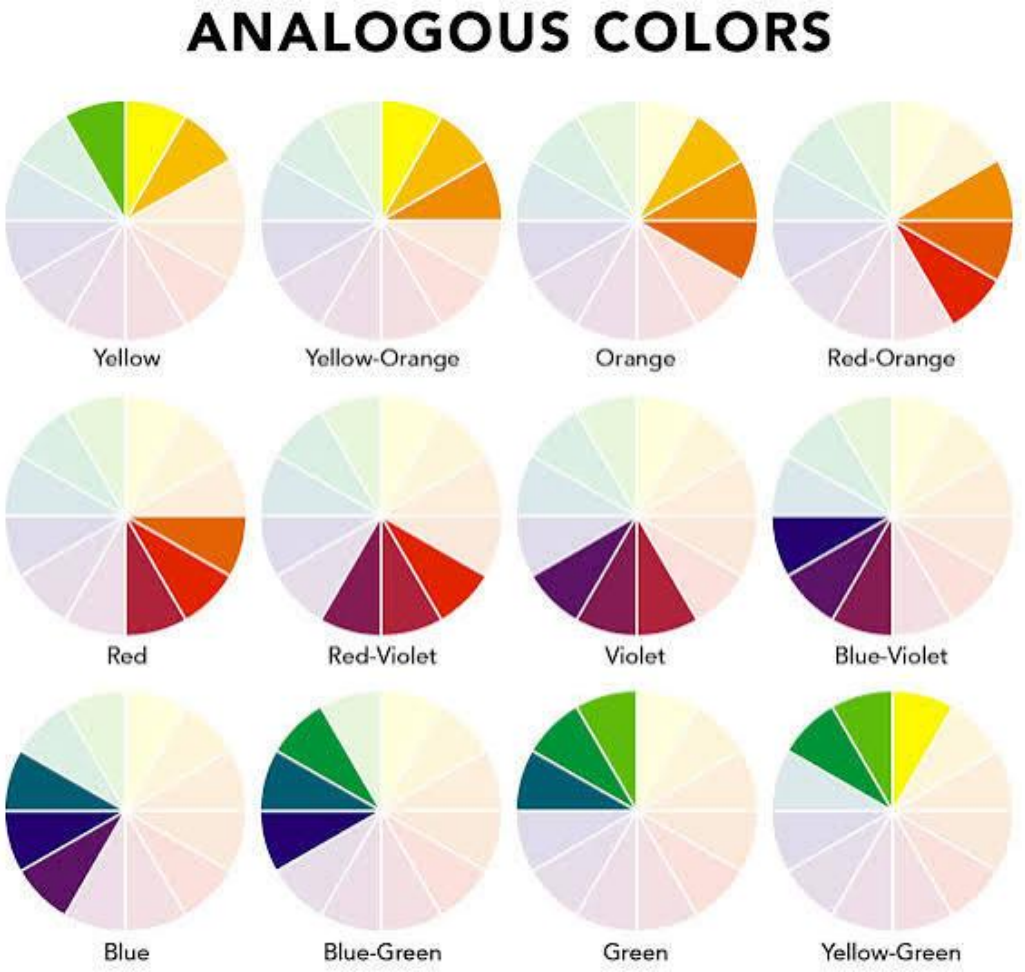
- **Monochromatic**
  - **Data is consecutive or varies**
  - **Uses one hue with different lightness or saturation**





II. Colour Theory

- **Analogous**
  - **Data more dynamic take on sequential**
  - **Show that categories are related.**
  - **Uses colours that are next to each other on the colour wheel.**



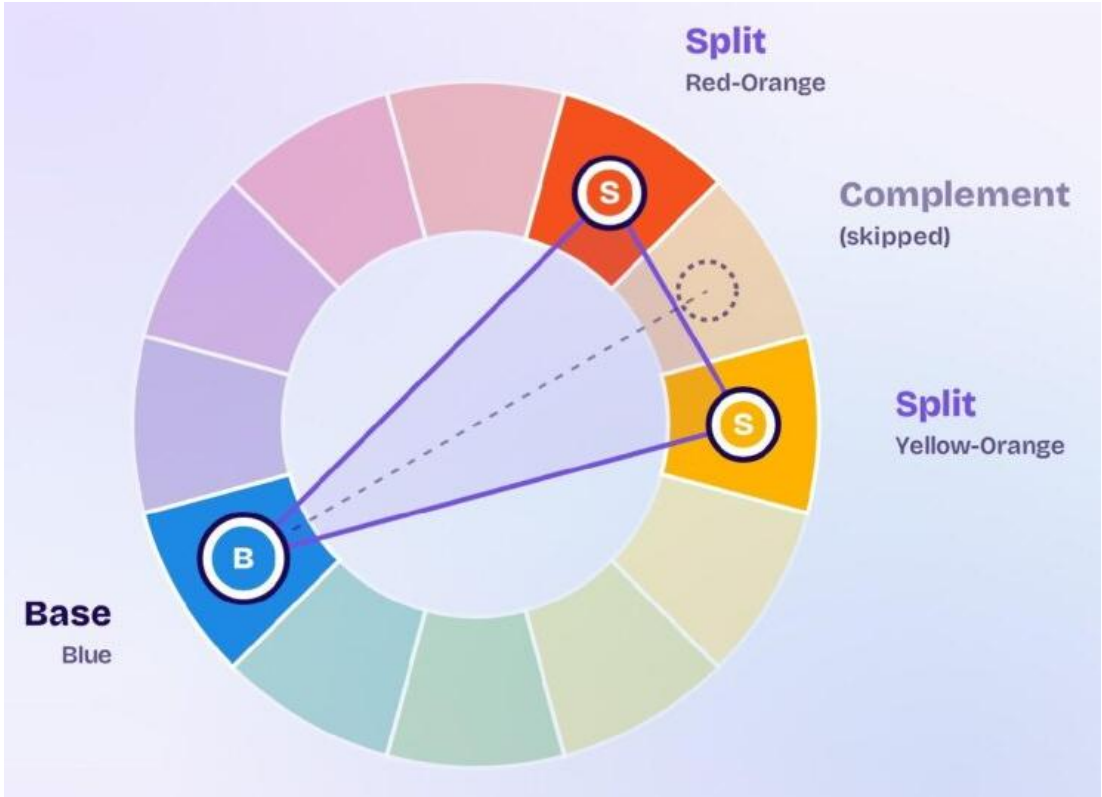
II. Colour Theory

- **Complementary**
  - **“positive” and “negative” influence losses or gains and**
  - **Emphasize differences or comparisons.**
  - **Uses colours opposite each other on the colour wheel.**



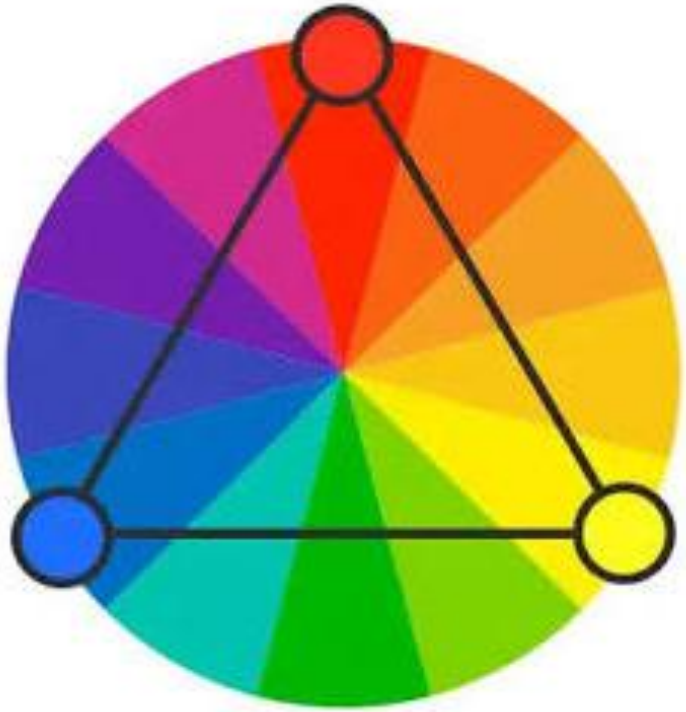
II. Colour Theory

- **Split-Complementary**
  - **Contrast, but less direct than complementary**
  - **Distinguish several categories.**
  - **Uses one colour plus the two colours beside its complementary colour.**



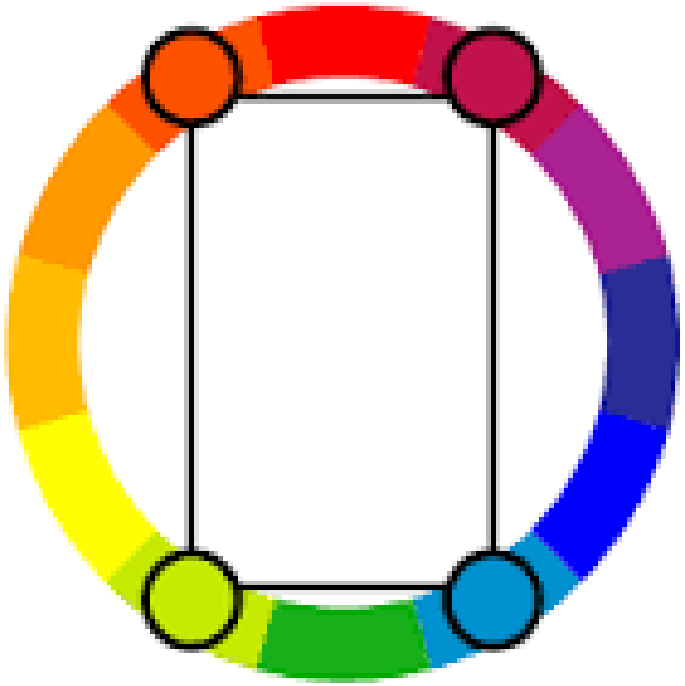
II. Colour Theory

- **Triad**
- **Distinguish multiple categories while maintaining visual balance.**
- **Uses three colours evenly spaced around the colour wheel.**
- **Make categories easy to identify**



II. Colour Theory

- **Tetradic**
- **Uses four colours based on two complementary pairs.**
- **Represent several distinct categories.**
- **Independent categories**



I. Colour Theory

❑ Maximising Colour Effectiveness in Data Visualization

- ❖ Using colour **deliberately and strategically so that it improves the communication of data**, rather than using colour simply to make a chart attractive.
- ❖ Use colour to help the viewer understand the data **quickly, accurately, and comfortably**.

I. Colour Theory

- **Colour Readability**
  - **Making sure colours in a data visualization are easy to see, distinguish, and understand by everyone.**
  - **Labels, symbols, patterns, shapes should use when necessary.**
- **Be consistent**
  - **Consistency in colour choice and its representation in visualisztions is key**
  - **Pattern recognition abilities, identify objects or entities represented**

Q&A



- Questions?
- Comments?
- Concerns?
- Suggestions?